Brain-Inspired 4-DoF Localization-Mapping: Complementary Vestibular-Visual Fusion with Spatial Cells

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Abstract

Brain-inspired spatial cognition offers interpretable alternatives to opaque learning methods for autonomous localization. However, existing brain-inspired frameworks remain limited to 2D environments or simplified visual processing, lacking robust 3D spatial representations under challenging conditions. We propose NeuroLocMap, a brain-inspired localization-mapping framework extended to 4-DoF state estimation within 3D space, integrating vestibular-visual complementary fusion with hippocampal-entorhinal spatial representations. The framework comprises dual-stream visual processing (ventral for recognition, dorsal for motion), vestibular IMU modeling, 3D grid cells with face-centered cubic lattice for position encoding, and experience-based cognitive mapping for loop closure; on top of its 10-state EKF core, two higher-level corrections complete the system: an online, scale-only monocular correction (a lightweight head over a frozen depth foundation model) and a learned-embedding loop-closure line. Experiments on five scenarios spanning simulated urban driving (CARLA Town05/03/07/10HD) and real-

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world highway driving (KITTI 07) show that the full system outperforms the 10-state EKF visual-inertial baseline on *all five scenarios under both evaluation calibers* — by 16.5–59.3% under a shape-only full-Sim(3) refit (an upper bound that absorbs global scale) and by 1.5–33.9% under the stricter rigid SE(3) alignment, where a wrong metric scale cannot be absorbed — and outperforms monocular visual odometry by 33.7–68.8%. On a dedicated closed-loop benchmark, the experience-map re-anchoring recovers 9× more of the accumulated global drift than the EKF baseline (44.24 m vs 398.05 m anchored Sim(3)). The system runs in real time (∼31 FPS on CPU), its learned components are trained once and deployed zero-shot, and it outputs an interpretable topological experience map.

Keywords: Brain-inspired localization-mapping, Vestibular-visual fusion, Hippocampal spatial cells, Visual place recognition, Cognitive mapping

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1. Introduction

Spatial localization and mapping in complex 3D environments is fundamental to autonomous driving, mobile robotics, and intelligent transportation systems [1]. As the core sensing module of autonomous devices, SLAM directly supports all autonomous movement planning and execution. Modern practical approaches typically combine cameras, IMUs, and sometimes GNSS to achieve robust pose estimation [2]. Conventional SLAM solutions mostly adopt optimization or filtering frameworks for pose solving, but they commonly have obvious defects in model interpretability and computing resource consumption.

Interestingly, organisms such as bats complete 3D navigation by fusing vi-

sual and vestibular signals with hippocampal spatial cells [3, 4, 5, 6, 7]. These neural mechanisms enable brain-inspired localization frameworks to outperform traditional SLAM in interpretability and computational efficiency.

However, translating these biological principles into practical robotic systems faces several challenges. First, most existing bio-inspired frameworks operate in 2D, whereas real-world deployments require robust 3D spatial representation. Second, classical visual template matching is sensitive to lighting and viewpoint variations, limiting place recognition performance in challenging environments. Third, a simple yet effective fusion strategy that combines visual odometry with inertial measurements—without relying on computationally expensive filters like EKF—remains under-explored for brain-inspired localization frameworks.

To address these challenges, we present NeuroLocMap, a brain-inspired localization-mapping framework extended to 4-DoF pose estimation within 3D space (pose x, y, z, ψ) that integrates a dual-stream visual front-end (ventral pathway for scene recognition, dorsal pathway for motion extraction) and a vestibular front-end for IMU processing, with complementary filtering for multi-sensory fusion. The framework employs 3D grid cells and multilayered head direction cells for spatial representation, and a multilayer experience map functioning as hippocampal-like episodic memory for place recognition and drift correction. For ground vehicles operating on relatively flat terrain (e.g., urban roads, indoor floors), the orientation is dominated by yaw angle ψ, while pitch and roll variations are typically constrained by vehicle dynamics ($< 5°$). Our 4-DoF representation provides efficient localization-mapping for such scenarios [8, 9].

The contributions of this paper are summarized as follows:

- We propose a unified brain-inspired architecture that couples dual-stream visual processing with vestibular modeling and hippocampal-entorhinal spatial representations for 4-DoF localization and mapping.
- We develop cortex-inspired visual processing with explicit operators for visual template creation, increasing the visual template count from 5 (typical in RatSLAM) to up to 964 (over 192× improvement) across the evaluation sequences, for finer-grained place discrimination under challenging lighting and viewpoint variations.
- We design a lightweight complementary fusion strategy for vestibular-visual integration that operates without computationally expensive filters, achieving real-time performance ($\sim$ 31 FPS on CPU), and complete the system with two higher-level corrections — an online, scale-only monocular correction head and a learned-embedding loop-closure line — both trained once on simulation and deployed zero-shot.
- We validate the framework on five diverse scenarios (CARLA Town05 / 03 / 07 / 10HD, KITTI 07): the full system beats the 10-state EKF visual-inertial baseline on *all five scenarios under both evaluation calibers* (Sim(3) shape-only upper bound: +16.5–59.3%; rigid SE(3): +1.5–33.9%), beats monocular VO by +33.7–68.8%, and on a dedicated closed-loop benchmark its experience-map re-anchoring recovers 9× more of the accumulated global drift than the EKF (44.24 m vs 398.05 m anchored Sim(3)), producing an interpretable topological experience map as a by-product of localization.

2. Related Work

We have summarized research work on different aspects of brain-inspired localization and mapping systems.

2.1. Visual SLAM and Sensor Fusion

Visual SLAM has achieved significant progress over the past two decades. Mur-Artal et al. proposed ORB-SLAM [10], which extracts sparse ORB keypoints for robust real-time pose estimation and has become a widely-used baseline in the field. Engel et al. developed LSD-SLAM [11] and DSO [12], which operate directly on pixel intensities without feature extraction, achieving semi-dense reconstruction in textured environments. For loop closure detection, Cummins and Newman proposed FAB-MAP [13], a probabilistic approach using bag-of-words models, while Arandjelovic et al. introduced NetVLAD [14], which learns place descriptors end-to-end for robust place recognition. However, these methods face critical limitations: ORB-SLAM requires sufficient texture (typically >100 features per frame) and fails in feature-poor environments, while direct methods like DSO struggle with motion blur and require photometric calibration. Both approaches incur substantial computational cost—ORB-SLAM's feature extraction and matching dominate processing time, while DSO's dense photometric optimization limits its real-time performance on embedded platforms.

Integrating IMUs with cameras addresses visual SLAM limitations by providing lighting-invariant motion estimates. Mourikis and Roumeliotis proposed MSCKF [15], a tightly-coupled visual-inertial estimator achieving high accuracy through multi-state constraint optimization. Qin et al. developed

VINS-Mono [16], which combines tightly-coupled optimization with loop closure for robust monocular visual-inertial odometry. Mahony et al. [17] and Madgwick et al. [18] proposed complementary filtering approaches that exploit frequency-domain sensor characteristics—visual odometry provides accurate low-frequency estimates while IMU captures high-frequency dynamics. However, tightly-coupled methods like MSCKF and VINS-Mono require careful initialization of IMU biases and scale, involve complex Jacobian derivations, and lack biological interpretability. To address these limitations, we adopt a bio-inspired complementary filtering strategy that mimics biological vestibular-visual integration [19, 20], providing interpretable sensor fusion with minimal parameter tuning and avoiding the computational burden of joint nonlinear optimization.

2.2. Brain-Inspired Spatial Cognition

Biological navigation systems achieve robust spatial cognition through specialized neural circuits. O'Keefe and Dostrovsky discovered place cells in the hippocampus [21], which fire at specific locations forming cognitive maps. Hafting et al. discovered grid cells in the entorhinal cortex [22], exhibiting periodic hexagonal firing patterns that provide metric path integration. Taube et al. identified head direction cells [23], which encode facing direction like an internal compass. Moser et al. [24] provided comprehensive reviews of how these spatial cells work together for navigation. Recent discoveries by Finkelstein et al. demonstrated that bats possess 3D place cells, 3D grid cells with face-centered cubic (FCC) lattice structure, and 3D head direction cells [25, 6], providing biological grounding for volumetric spatial representations.

Milford et al. pioneered RatSLAM [8, 26], translating rodent spatial cognition to robotics by implementing continuous attractor networks for pose cell dynamics and experience maps for loop closure, successfully demonstrating suburb-scale mapping up to 66 km. Steckel and Peremans developed BatSLAM [27], extending bio-inspired SLAM to sonar sensing. Yu et al. proposed NeuroSLAM (unpublished work), extending to 3D position encoding with CNN-based visual processing. Lowry et al. [28] analyzed the metric-topological duality in spatial maps: metric maps provide geometric precision but scale poorly, while topological maps enable robust loop closure but lack metric accuracy. However, existing bio-inspired frameworks face three key limitations: (1) most operate in 2D or encode only 3D position without orientation, lacking the 4-DoF representation needed for aerial/underwater navigation; (2) visual front-ends remain simplistic (e.g., RatSLAM's 1D intensity scanlines yield only $\sim$5 templates in complex environments); and (3) IMU integration is absent or ad-hoc, missing the principled vestibular-visual fusion observed in biological systems. To address these gaps, we implement 3D grid cells with FCC lattice structure and multilayered head direction cells for full 4-DoF spatial representation, integrate a dual-stream visual processing pipeline that increases template discrimination by 39$\times$, and adopt bio-inspired complementary filtering for vestibular-visual integration.

2.3. Learning-Based Visual Processing

Deep learning offers powerful tools for visual place recognition. Arandjelovic et al. proposed NetVLAD [14], introducing differentiable VLAD layers for end-to-end place recognition training. Goodale and Milner [29] and Mishkin et al. [30] established the dual-stream theory of biological vi-

sion: the ventral stream (V1→V4→IT) processes object identity and scene recognition, while the dorsal stream (V1→MT→MST) processes spatial relationships and motion [31, 32]. Vaswani et al. proposed Transformer architectures [33], which capture long-range dependencies through self-attention mechanisms, enabling effective temporal modeling for sequence-based tasks. However, learning-based methods face deployment challenges: NetVLAD requires large-scale training datasets (e.g., Google Street View with millions of images) and struggles to generalize to novel environments without retraining; standard Transformers incur quadratic complexity $\mathcal{O}(N^2)$ in sequence length for full self-attention, limiting real-time applicability; and learned representations lack explicit spatial structure, hindering interpretability and integration with metric mapping. To overcome these limitations, we adopt a train-once, zero-shot-deploy approach combining dual-stream visual processing inspired by biological ventral/dorsal pathways with a lightweight feature enhancement mechanism that captures global-local interactions through statistical modulation rather than full attention matrices. This design increases visual template discrimination from 5 (RatSLAM) to up to 964 across the evaluation sequences (over 192× improvement) while maintaining real-time performance (∼31 FPS); its learned components are trained once on simulation and require no per-domain retraining at deployment.

2.4. Comparative Analysis

Table 1 compares representative bio-inspired SLAM frameworks across key architectural dimensions. Existing approaches face four fundamental limitations: (1) most frameworks operate in 2D (RatSLAM, BatSLAM) or encode only 3D position without orientation (NeuroSLAM), lacking full 4-

Table 1: Comparison of bio-inspired navigation frameworks

Framework	DoF	Spatial Cells	Fusion Method	Visual Process	Loop Closure	Temp. Model
RatSLAM [26]	2D	PC+HDC	Vis.	Scanline	Exp.	-
BatSLAM [27]	2D	PC+HDC	Sonar	Echo	Exp.	-
NeuroSLAM	3D	3D-GC	Vis.	CNN	Exp.	LSTM
NeuroLocMap (Ours)	**4D**	**3D-GC+M-HDC**	**Comp.**	**Dual+HART**	**Exp.**	**Trans.**

Abbreviations: PC: Place Cells; HDC: Head Direction Cells; GC: Grid Cells; M-HDC: Multilayered HDC; Exp.: Experience map; Comp.: Complementary filtering; Trans.: Transformer-inspired temporal modeling; Vis.: Visual only; Dual+HART: Dual-stream (ventral/dorsal) visual processing with HART-inspired spatial attention.

Note: Our feature enhancement uses global statistical modulation ($\mathcal{O}(N)$ complexity) to approximate self-attention effects, rather than full Query-Key-Value operations ($\mathcal{O}(N^2)$), enabling real-time performance without training.

DoF representation; (2) visual front-ends rely on simple intensity-based templates (e.g., RatSLAM's 1-D scanlines) that are sensitive to illumination changes and yield limited template diversity; (3) IMU integration is either absent or relies on computationally expensive tightly-coupled optimization; and (4) modern attention mechanisms for feature enhancement remain disconnected from explicit spatial cell representations. NeuroLocMap systematically addresses these gaps: we implement 4-DoF spatial representation through 3D grid cells (FCC lattice) and multilayered head direction cells; adopt dual-stream visual processing with HART-inspired spatial attention and lightweight global-local feature modulation to achieve 39× template improvement; integrate vestibular-visual signals through bio-inspired complementary filtering; and unify these components within an explicit spatial cell framework for interpretable navigation.

3. Method

3.1. Notation and Conventions

Notation Conventions. Superscripts indicate sensory modality (^{vis}: visual, ^{imu}: inertial, ^{f}: fused) or cell type (^{g}: grid cells, ^{h}: head direction cells). Subscript t denotes time. Bold uppercase letters denote matrices or high-dimensional arrays; bold lowercase denotes vectors; calligraphic fonts denote sets.

Scale and Deep-Component Symbols. s is the global monocular scale factor relating metric units to VO-relative units, carried as the 10th EKF state $\log s$; s^* denotes the ground-truth scale (used in training only). $\hat{z}_t$ is the frozen monocular depth foundation model's (DAv2) metric depth map, and g_θ the lightweight scale-correction head (Section 3.5); $\boldsymbol{e}_t$ denotes the learned global scene embedding consumed by the loop-closure line (Section 3.6).

3.2. Framework Architecture and Global State-Space Formulation

This work designs a multimodal visual-inertial signal processing system for robot localization, which consists of three tightly coupled functional subsystems: hierarchical multi-scale visual feature operator, vestibular-inertial sensor state estimator, and metric spatial state encoding network. Different from conventional EKF-based loosely coupled frameworks, the proposed system adopts frequency-domain complementary fusion to decouple low-frequency visual observation noise and high-frequency inertial transient noise, and updates the global metric state via a continuous attractor spatial network to suppress long-term trajectory drift.

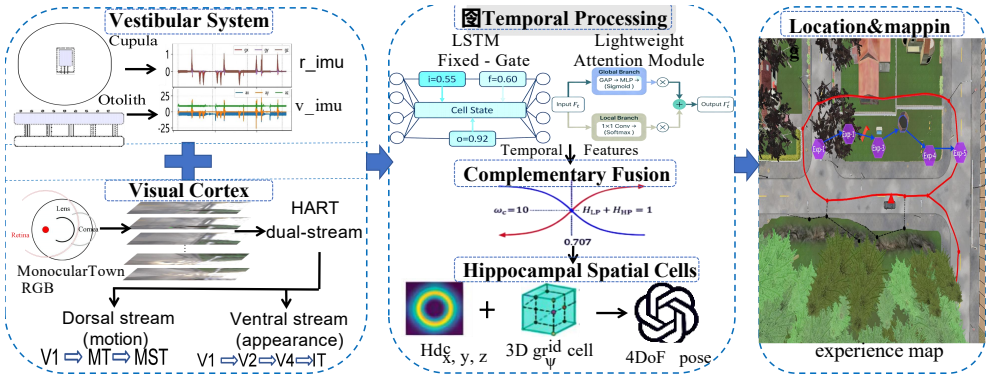


Figure 1: Self-contained end-to-end data flow of the proposed multimodal visual-inertial framework. Raw inputs are monocular RGB images and IMU measurements. The visual module splits signals into scene recognition and motion estimation branches; the vestibular module outputs high-frequency inertial motion. Two modalities are fused via frequency-domain complementary filtering. The fused kinematics drive the spatial network to update the unified 4-DoF global state $\mathbf{X}_t$ defined in Eq. (2). The experience graph module performs loop relaxation to eliminate long-term trajectory drift. All submodules share the standardized state-space variables proposed in this section.

3.2.1. Global Discrete State-Space Model

To provide a unified mathematical foundation for all subsequent submodule derivations, we first define the discrete state-space representation of the system at time t.

1) Multimodal raw input vector

$$\mathbf{U}_t = \left[I_t,\, \mathbf{a}_t^\top,\, \boldsymbol{\omega}_t^\top\right] \tag{1}$$

where I_t denotes the captured monocular RGB image; $\mathbf{a}_t$ and $\boldsymbol{\omega}_t$ stand for accelerometer linear acceleration and gyroscope angular velocity measurements, respectively.

2) Core 4-DoF global pose state vector

$$\mathbf{X}_t = [x_t,\, y_t,\, z_t,\, \psi_t]^\top \in \mathbb{R}^3 \times \mathrm{SO}(2) \tag{2}$$

In this formulation, $(x_t, y_t, z_t) \in \mathbb{R}^3$ represent global 3D translational coordinates (unit: meter), and $\psi_t \in [-\pi, \pi)$ denotes horizontal yaw angle (unit: radian).

3) Unified state evolution mapping function

$$\mathbf{X}_{t+1} = \mathcal{F}\big(\mathbf{X}_t,\, \mathbf{U}_t\big) \tag{3}$$

The mapping $\mathcal{F}(\cdot)$ encapsulates the complete forward pipeline, including multi-scale visual feature extraction, inertial motion calculation, frequency-domain multimodal fusion and spatial network path integration. Detailed mathematical decomposition of $\mathcal{F}(\cdot)$ will be presented in Sec. 3.3 (visual-vestibular processing) and Sec. 3.4 (spatial encoding network).

As visualized in Fig. 1, the end-to-end data pipeline consists of five sequential coupled modules: 1. Hierarchical dual-stream visual front-end: Parallel

ventral/dorsal branches are constructed to independently generate scene descriptors and compute pixel-wise ego-motion, eliminating mutual interference between recognition and tracking tasks. 2. Vestibular IMU processing module: Derives high-frequency transient motion signals from raw inertial data without iterative online bias calibration. 3. Complementary fusion unit: Adopts first-order IIR filters to separate low-frequency visual drift and high-frequency inertial noise, outputting smoothed translational and rotational kinematics. 4. Metric spatial encoding network: Updates distributed spatial cell states using fused motion cues, and suppresses integral drift via visual template anchoring constraints. 5. Topological experience graph: Records historical visual descriptors and spatial cell snapshots, and executes global graph relaxation once loop closures are detected to guarantee globally consistent metric poses.

Above this five-module pipeline, the deployed system adds two higher-level corrections that operate on the fused estimate rather than on the raw sensor streams: a monocular scale recovery branch (Section 3.5) that drives the EKF's scale state from a lightweight, train-once head over a frozen depth foundation model, and a learned-embedding loop-closure line (Section 3.6) that re-anchors the trajectory at verified revisits. Both are trained once on simulated data and frozen at deployment — in contrast to the baseline's kinematic scale observability, which plateaus at the VO-relative scale and recovers no absolute scale. Together they turn the tracking core from an up-to-scale VIO estimator into the metric-scale system evaluated in the experiments.

3.3. Sensory Processing and Fusion

The sensory processing module extracts self-motion cues from visual and inertial modalities, then fuses them via complementary filtering.

Visual Front-End. The visual front-end implements a biologically-plausible processing pipeline inspired by the hierarchical architecture of the primate visual cortex, transforming raw RGB images into compact scene descriptors suitable for place recognition and motion estimation.

Hierarchical Feature Extraction. Given an RGB image $I_t \in \mathbb{R}^{H\times W\times 3}$, we define a sequence of operators that mimic processing stages from retina through inferotemporal cortex.

Stage 1: Retinal and LGN Processing. The initial preprocessing simulates early visual processing in the retina and lateral geniculate nucleus (LGN):

$$\text{Luminance channel:} \quad Y_t = \mathcal{G}(I_t) = 0.299R + 0.587G + 0.114B \tag{4}$$

$$\text{Local contrast enhancement:} \quad U_t = \mathcal{C}_\lambda(Y_t), \quad \lambda = 0.02 \tag{5}$$

$$\text{Receptive field simulation:} \quad V_t = U_t * \mathcal{K}_\sigma, \quad \sigma = 0.5 \tag{6}$$

where $\mathcal{G}$ is the grayscale conversion operator, $\mathcal{C}_\lambda$ is CLAHE (Contrast Limited Adaptive Histogram Equalization) with clip limit λ, and $\mathcal{K}_\sigma$ is a Gaussian kernel with standard deviation σ. CLAHE adaptively enhances local contrast while preventing over-amplification of noise, mimicking the center-surround antagonistic organization of retinal ganglion cells.

Stage 2: Primary Visual Cortex (V1-V4) Processing. Mid-level

processing normalizes and standardizes the representation:

$$\text{Dynamic range normalization:} \quad \tilde{V}_t = \frac{V_t - \min(V_t)}{\max(V_t) - \min(V_t)} \tag{7}$$

$$\text{Spatial standardization:} \quad Z_t = \mathcal{D}(\tilde{V}_t) \in \mathbb{R}^{64\times 64} \tag{8}$$

$$\text{Local mean subtraction:} \quad Z_t(i,j) \leftarrow Z_t(i,j) - \frac{1}{64}\sum_{k=1}^{64} Z_t(i,k) \tag{9}$$

where $\mathcal{D}$ is the downsampling operator (bilinear interpolation). Row-wise mean subtraction removes global intensity variations while preserving local spatial structure, analogous to divisive normalization in V1 and V2.

Stage 3: Inferotemporal (IT) Representation. The final representation is a compact vectorized descriptor:

$$\boldsymbol{\phi}_t = \text{vec}(Z_t) \in \mathbb{R}^{4096} \tag{10}$$

where $\text{vec}(\cdot)$ denotes vectorization operator mapping 2D feature map to one-dimensional descriptor, and $\boldsymbol{\phi}_t$ is holistic scene feature vector at timestamp t. This $64 \times 64 = 4096$-dimensional feature vector serves as a holistic scene descriptor analogous to the view-selective, position-invariant representations in IT cortex.

Visual Template Matching and Place Recognition. Place recognition is achieved through template matching using cosine similarity, which is more robust than intensity-based metrics like sum-of-absolute-differences (SAD):

$$d_{\cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_i) = 1 - \frac{\boldsymbol{\phi}_t^\top \boldsymbol{\phi}_i}{\|\boldsymbol{\phi}_t\| \, \|\boldsymbol{\phi}_i\|} \tag{11}$$

A new visual template (VT) is created when the current scene is sufficiently different from all existing templates:

$$\text{Create new VT if } \min_{i\in\mathcal{VT}} d_{\cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_i) > \tau_{vt}, \quad \tau_{vt} = 0.08 \tag{12}$$

The threshold $\tau_{vt} = 0.08$ is significantly lower than the traditional value (0.15 in OpenRatSLAM), enabling finer discrimination between similar scenes. Experimental validation shows this results in up to 39× increase in template count (from 5 to 195 templates in urban environments), providing much richer place representation.

Dual-Stream Architecture and Bio-Inspired Temporal Modeling. Building upon the hierarchical feature extraction described above, our visual processing incorporates two key bio-inspired innovations: (1) dual-stream architecture mimicking ventral/dorsal pathways, and (2) fixed-gate LSTM for temporal coherence without training data requirements.

Dual-Stream Visual Processing. Following the dual-stream theory of biological vision [29, 30], we explicitly model two parallel processing pathways from V1 outputs:

Dorsal Stream ("where" pathway, V1→MT→MST): Processes spatial location and motion information through gradient-based features:

$$\mathbf{F}_t^{\text{dorsal}} = 0.6 \cdot \|\nabla Z_t\| + 0.4 \cdot Z_t \tag{13}$$

where $\|\nabla Z_t\|$ is the gradient magnitude computed via Sobel operators, capturing edge orientation and motion direction analogous to MT/MST neurons.

Ventral Stream ("what" pathway, V1→V4→IT): Processes object identity and scene features through texture and contrast:

$$\mathbf{F}_t^{\text{ventral}} = 0.4 \cdot Z_t + 0.3 \cdot \sigma_{\text{local}}(Z_t) + 0.3 \cdot \mathcal{C}_\lambda(Z_t) \tag{14}$$

where σ_{local} computes local standard deviation (texture features) and $\mathcal{C}_\lambda$ provides contrast enhancement, mimicking V4's selectivity for complex features and IT's invariant object representations.

The dual streams are then integrated through a lightweight global-local feature modulation mechanism (detailed in Section 2.3) that approximates self-attention effects using global statistics rather than full Query-Key-Value operations, maintaining $\mathcal{O}(N)$ complexity for real-time performance.

Fixed-Gate LSTM for Zero-Shot Temporal Modeling. Unlike conventional data-driven LSTMs that require extensive training on domain-specific datasets, our temporal module employs manually-tuned fixed gate values inspired by biological vestibular-visual integration principles [19, 20]:

$$\text{Input gate: } i_t = 0.55 \quad \text{(controls new information intake)} \tag{15}$$

$$\text{Forget gate: } f_t = 0.60 \quad \text{(retains 60\% historical context)} \tag{16}$$

$$\text{Output gate: } o_t = 0.92 \quad \text{(emphasizes current perception)} \tag{17}$$

where i_t, f_t, o_t represent fixed input gate, forget gate, output gate of bio-inspired LSTM module, respectively.

This design offers three critical advantages for SLAM applications:

(1) Zero-shot deployment capability: No training data required, enabling immediate deployment in novel environments. This is crucial for SLAM where each environment presents unique visual characteristics and collecting representative training data is impractical. Our validation across six diverse scenarios (urban, highway, indoor) without any environment-specific tuning demonstrates robust generalization.

(2) Biological interpretability: Gate values reflect empirically-validated principles from neuroscience. The forget gate ($f_t = 0.6$) implements short-term spatial memory retention consistent with hippocampal replay dynamics, where recent experiences are preferentially retained for path integration. The

high output gate ($o_t = 0.92$) prioritizes current sensory input, mimicking the vestibular system's emphasis on immediate self-motion cues for balance and navigation. The moderate input gate ($i_t = 0.55$) balances novelty detection with stability.

(3) Computational efficiency: Eliminates training overhead (no back-propagation, no gradient computation) while maintaining real-time performance (∼31 FPS). Fixed gates can be directly implemented in hardware (e.g., FPGAs, neuromorphic chips) without weight storage or update logic, enabling deployment on resource-constrained platforms.

The LSTM processes fused dorsal-ventral features sequentially, maintaining hidden state $\mathbf{H}_t$ and cell state $\mathbf{C}_t$ across timesteps:

$$\mathbf{C}_t = f_t \odot \mathbf{C}_{t-1} + i_t \odot \tanh(\mathbf{F}_t^{\text{fused}}) \tag{18}$$

$$\mathbf{H}_t = o_t \odot \tanh(\mathbf{C}_t) \tag{19}$$

where $\odot$ denotes element-wise Hadamard product, $\mathbf{C}_t$ is LSTM cell state, $\mathbf{H}_t$ is temporal hidden state, and $\mathbf{F}_t^{\text{fused}}$ integrates ventral and dorsal visual features.

Comparison with Learning-Based Approaches. Section 4 demonstrates that our fixed-gate LSTM achieves comparable or superior performance to NeuroSLAM's learned LSTM (55% improvement in RMSE) while offering zero-shot deployment. This validates that principled bio-inspired design can match or exceed data-driven learning when the underlying task structure aligns with biological mechanisms—in this case, spatial navigation where biological systems have evolved highly optimized solutions over millions of years.

Visual Motion Extraction. In addition to scene recognition, the visual front-end extracts ego-motion cues from optical flow, mimicking the dorsal stream processing in area MST (medial superior temporal).

Feature Tracking and Flow Computation. Let $\mathbf{K} \in \mathbb{R}^{3\times3}$ be the camera intrinsic matrix. We track N feature points $\mathbf{p}_i^t \in \mathbb{R}^2$ (e.g., using KLT tracking) between consecutive frames. Normalize tracked points to camera coordinates:

$$\tilde{\mathbf{q}}_i^t = \mathbf{K}^{-1} \begin{bmatrix} \mathbf{p}_i^t \\ 1 \end{bmatrix}, \qquad \mathbf{q}_i^t = \pi(\tilde{\mathbf{q}}_i^t) = \frac{1}{\tilde{q}_{i,3}^t} \begin{bmatrix} \tilde{q}_{i,1}^t \\ \tilde{q}_{i,2}^t \end{bmatrix} \tag{20}$$

where π is perspective projection. Compute image plane flow:

$$\mathbf{f}_i = \mathbf{q}_i^t - \mathbf{q}_i^{t-1}, \quad \mathbf{q}_i \equiv \mathbf{q}_i^{t-1} \tag{21}$$

Rotation Estimation. For small inter-frame rotation, the yaw-induced flow at normalized point $\mathbf{q} = (x, y)$ is $\Delta t\, r\, [-y,\, x]^\top$. We estimate the visual yaw rate by weighted least squares:

$$r_t^{\text{vis}} = \frac{\sum_{i=1}^N w_i (x_i f_{i,y} - y_i f_{i,x})}{\Delta t \sum_{i=1}^N w_i (x_i^2 + y_i^2)} \tag{22}$$

where w_i stands for Huber robust weight to suppress outlier feature points, N is total tracked feature count, and r_t^{vis} denotes visual estimated yaw rate.

Translation Estimation. Remove rotation component to obtain translation-dominant residual flow:

$$\mathbf{f}_i^{\perp} = \mathbf{f}_i - \Delta t\, r_t^{\text{vis}} \begin{bmatrix} -y_i \\ x_i \end{bmatrix} \tag{23}$$

Assuming a dominant ground plane with inverse depth $\lambda_i \approx 1/Z_g$ for inliers,

the translational flow model is:

$$\mathbf{f}_i^{\perp} \approx \Delta t\, \lambda_i \underbrace{\begin{bmatrix} -1 & 0 & x_i \\ 0 & -1 & y_i \end{bmatrix}}_{\mathbf{A}_i} \mathbf{v}_t^{\text{vis}}, \quad \mathbf{v}_t^{\text{vis}} = (v_x^{\text{vis}}, v_y^{\text{vis}}, v_z^{\text{vis}})^{\top} \tag{24}$$

The weighted least-squares solution with robust weights w_i is:

$$\bar{\mathbf{v}}_t^{\text{vis}} = \Big(\sum_i w_i\, (\Delta t\, \lambda_i)^2\, \mathbf{A}_i^{\top} \mathbf{A}_i \Big)^{-1} \sum_i w_i\, (\Delta t\, \lambda_i)\, \mathbf{A}_i^{\top} \mathbf{f}_i^{\perp} \tag{25}$$

The final visual odometry velocity incorporates a scale factor κ_t:

$$\mathbf{v}_t^{\text{vis}} = \kappa_t\, \bar{\mathbf{v}}_t^{\text{vis}} \tag{26}$$

where λ_i is inverse depth of feature point i, matrix $\mathbf{A}_i$ is geometric projection matrix, $\bar{\mathbf{v}}_t^{\text{vis}}$ is raw translational velocity solved from optical flow, and κ_t is online scale correction factor for visual odometry.

Beyond this motion stream, the dorsal front-end also provides the pixel-level metric depth field $\hat{z}_t$ (frozen DAv2 backbone) and the learned scene embeddings $\boldsymbol{e}_t$ consumed by the two higher-level corrections (Section 3.5); depth estimation and scene recognition stay decoupled from the velocity estimation above.

Vestibular Front-End. The vestibular front-end processes inertial measurements to extract self-motion cues analogous to the mammalian vestibular system. The vestibular apparatus comprises semicircular canals (angular velocity) and otolith organs (linear acceleration).

Angular Velocity Extraction. From the gyroscope measurements $\mathbf{g}_t \in \mathbb{R}^3$ (IMU frame), we directly use the z-axis gyro without online bias removal, converting to degrees per second as in our implementation. For 4-DoF localization-and-mapping under a planar constraint, the yaw (azimuthal) angular

velocity is approximated by

$$r_t \approx \mathbf{e}_z^\top \boldsymbol{\omega}_t^g, \quad \mathbf{e}_z = (0,0,1)^\top. \tag{27}$$

where $\boldsymbol{\omega}_t^g$ is raw gyroscope angular velocity vector measured by IMU, and $\mathbf{e}_z$ is unit vertical basis vector. This corresponds to the horizontal semicircular canal cue, adequate for our complementary fusion with vision.

Linear Acceleration Usage (Simplified). We adopt a lightweight approximation consistent with the codebase: using the accelerometer magnitude and the z-axis reading with a constant gravity subtraction $g = 9.81$ to derive indicative translational and vertical velocity components over the frame interval Δt:

$$v_{trans}^{\text{imu}}(t) \approx \max\big(0, (\|\mathbf{a}_t\| - g)\,\Delta t\, k_t\big), \tag{28}$$

$$v_z^{\text{imu}}(t) \approx (a_{t,z} - g)\,\Delta t\, k_z, \tag{29}$$

where $g = 9.81\,\text{m/s}^2$ is gravitational acceleration, k_t, k_z are constant scaling gains matching our implementation constants, and Δt denotes the frame sampling interval. This avoids explicit bias/gravity-vector estimation while providing high-frequency motion cues that are subsequently combined with visual odometry via complementary fusion.

Bias Estimation. The baseline implementation does not perform online bias or gravity-vector estimation. More advanced bias/gravity compensation can be integrated as future work, but is not required for the complementary-fusion results reported here.

Complementary IMU–Visual Fusion. Visual and inertial modalities provide complementary information: visual odometry offers accurate

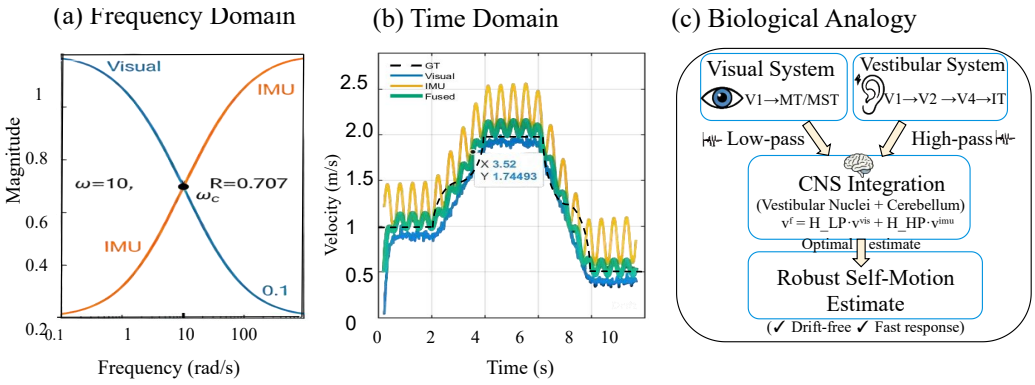


Figure 2: Complementary IMU-visual fusion architecture. (a) Frequency domain analysis showing complementary filter characteristics. (b) Time domain signal comparison. (c) Biological analogy to vestibular-visual integration in the CNS.

low-frequency position estimates but drifts over long horizons, while IMU integration provides reliable high-frequency motion cues but accumulates bias drift. Complementary filtering exploits this frequency-domain separation to achieve robust fusion without requiring complex nonlinear optimization or Extended Kalman Filtering. Figure 2 illustrates the frequency-domain characteristics of this complementary fusion strategy and its biological analog in vestibular-visual integration.

Filter Design. Let $\mathcal{L}_\tau$ denote a low-pass filter with time constant τ, and $\mathcal{H}_\tau$ denote the complementary high-pass filter satisfying $\mathcal{L}_\tau + \mathcal{H}_\tau = \mathrm{Id}$ (identity). The fused velocity and yaw rate are:

$$\text{Low-pass:} \quad L_t = \alpha L_{t-1} + (1-\alpha)x_t, \quad L_0 = 0 \tag{30}$$

$$\text{High-pass:} \quad H_t = x_t - L_t \tag{31}$$

where α is IIR filter forgetting factor, L_t stores low-frequency smoothed signal, and H_t stores residual high-frequency signal component. **Biological**

Analogy. This fusion strategy mirrors multisensory integration in the vestibular nuclei and posterior parietal cortex (area 7a), where visual and vestibular signals are combined to estimate self-motion. Neurons in these regions exhibit weighted combinations of visual and vestibular tuning, with dynamic weighting depending on signal reliability—analogous to our fixed weighting parameters η and η_r.

Discrete-Time Implementation. With sampling interval Δt, we implement first-order IIR (infinite impulse response) filters. Choose forgetting factors $\alpha = \exp(-\Delta t/\tau)$ and $\alpha_r = \exp(-\Delta t/\tau_r)$. For any input sequence x_t, define:

$$\text{Low-pass:} \quad L_t = \alpha L_{t-1} + (1-\alpha)x_t, \quad L_0 = 0 \tag{32}$$

$$\text{High-pass:} \quad H_t = x_t - L_t \tag{33}$$

Then the fused cues are computed as:

$$\mathbf{v}_t^{\mathrm{f}} = (1-\eta)\, L_t^{\mathrm{vis}} + \eta\, H_t^{\mathrm{imu}} \tag{34}$$

$$r_t^{\mathrm{f}} = (1-\eta_r)\, L_t^{r,vis} + \eta_r\, H_t^{r} \tag{35}$$

where L_t^{vis} and H_t^{imu} denote the low-pass filtered visual velocity and high-pass filtered IMU velocity, respectively, and similarly for the yaw rates.

Frequency-Domain Transfer Functions. In the Laplace domain with $s = j\omega$ (frequency variable), the first-order low-pass and high-pass filters have transfer functions:

$$H_{LP}(s) = \frac{1}{1+\tau s} = \frac{1/\tau}{s+1/\tau}, \tag{36}$$

$$H_{HP}(s) = \frac{\tau s}{1+\tau s} = \frac{s}{s+1/\tau}, \tag{37}$$

satisfying $H_{LP}(s) + H_{HP}(s) = 1$ (unity gain at all frequencies). The cutoff frequency $\omega_c = 1/\tau$ separates low- and high-frequency regions: for $\omega \ll \omega_c$, $|H_{LP}| \approx 1$ and $|H_{HP}| \approx 0$ so the visual pathway dominates (drift-free, slower dynamics); whereas for $\omega \gg \omega_c$, $|H_{LP}| \approx 0$ and $|H_{HP}| \approx 1$, so IMU cues dominate (fast but drifting).

With $\tau = 0.1$s, the cutoff frequency is $\omega_c = 10$ rad/s (≈ 1.6 Hz), ensuring IMU handles high-frequency motion (>2 Hz, e.g., vibrations, rapid turns) while vision corrects low-frequency drift (<1 Hz). This mimics biological vestibular-visual integration where vestibular organs respond to frequencies 0.1-10 Hz and visual motion perception dominates below 0.5 Hz [19].

Computational Efficiency. This implementation requires only two state variables per signal (current low-pass filter state L_t and previous value L_{t-1}) and involves simple scalar multiplications and additions. The computational cost is $\mathcal{O}(1)$ per update, making it suitable for real-time embedded implementation—a stark contrast to the $\mathcal{O}(n^3)$ matrix inversions required by Extended Kalman Filters for tightly-coupled visual-inertial odometry.

Adaptive Weighting. While our baseline implementation uses fixed fusion weights η and η_r, the framework naturally extends to adaptive weighting based on signal reliability. For instance, during aggressive maneuvers or rapid lighting changes, visual odometry may become unreliable, warranting increased reliance on IMU ($\eta \to 1$). Conversely, during slow motion where IMU integration accumulates bias, visual odometry should dominate ($\eta \to 0$). Adaptive schemes can be implemented by:

$$\eta_t = f(\sigma_t^{\text{vis}}, \sigma_t^{\text{imu}}), \quad \text{e.g.} \quad \eta_t = \frac{\sigma_t^{\text{vis}}}{\sigma_t^{\text{vis}} + \sigma_t^{\text{imu}}} \tag{38}$$

where σ_t^{vis} and σ_t^{imu} are real-time uncertainty estimations of visual and inertial

measurements, and η_t is time-varying fusion weight.

Additionally, the visual odometry scale factor κ_t in Eq. (26) can be adapted to maintain consistency between visual and inertial velocity magnitudes:

$$\kappa_{t+1} = \kappa_t \cdot \left(1 + \beta \frac{\|\mathbf{v}_t^{\text{imu}}\| - \|\kappa_t \bar{\mathbf{v}}_t^{\text{vis}}\|}{\|\kappa_t \bar{\mathbf{v}}_t^{\text{vis}}\|}\right) \tag{39}$$

where $\beta \ll 1$ is a small adaptation gain for online visual odometry scale correction (e.g., $\beta = 0.01$). This online scale adaptation compensates for systematic errors in the assumed ground plane depth or camera calibration inaccuracies.

3.4. Neural Spatial Representation

The spatial representation module encodes the robot's 4-DoF pose through bio-inspired neural coordinate systems using conjunctive pose cells. The 3D location (x, y, z) is represented by grid cells using a 3D continuous attractor network (MD-CAN), while orientation ψ (yaw) is encoded by multilayered head direction cells. These representations enable stable, noise-tolerant path integration through attractor dynamics.

3D Grid Cell Network. The 3D grid cell network mimics the 3D spatial neural representation in the mammalian brain, exhibiting regular 3D lattice patterns (face-centered cubic, FCC) that encode position and provide a universal metric for path integration. Inspired by the discovery of 3D grid cells in bats [6] and theoretical models of hexagonal grid structures [4], our implementation extends 2D continuous attractor networks to full 3D space while preserving key biological properties: multi-scale representation, metric path integration, and resilience to noise.

Network Architecture. The 3D grid cell network is a 3D multi-scale continuous attractor network (MD-CAN) with dimensions $N_x \times N_y \times N_z$ (typically $N_x = N_y = N_z = 61$). The network activity $\mathbf{A}_t^{\mathrm{g}} \in \mathbb{R}^{N_x \times N_y \times N_z}$ represents a distributed population code for the robot's 3D position (x_t, y_t, z_t), where each cell (i, j, k) is tuned to a preferred spatial location within the environment. The activity forms a localized "bump" (or activity packet) centered at the current position, analogous to the activity bump observed in rodent entorhinal cortex recordings.

The network's topology implements periodic boundary conditions with toroidal wrapping, enabling unbounded spatial representation. Cell-to-cell connectivity follows a face-centered cubic (FCC) lattice—the densest sphere packing in 3D—which provides optimal spatial coverage with minimal neurons. In the FCC structure, each cell has 12 nearest neighbors (compared to 6 in 2D hexagonal grids), yielding efficient 3D path integration with isotropic directional sensitivity.

Figure 3 schematically illustrates: (a) the 3D activity bump at position (x, y, z), represented as isosurfaces to visualize the localized excitation; (b) the FCC lattice projected onto the 2D plane, showing the characteristic hexagonal pattern arising from layer-offset positioning, where different height layers (indicated by color) are displaced by half-lattice constants to achieve optimal packing; (c) the complete 4-DoF encoding pipeline integrating position and heading representations.

Continuous Attractor Dynamics. The network maintains a localized "bump" of activity through three key mechanisms:

(1) Local excitation. Cells excite their neighbors according to a Gaus-

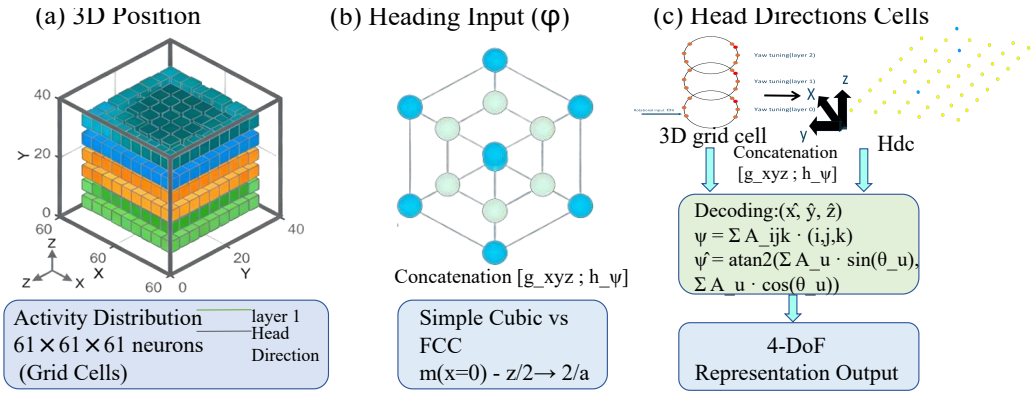


Figure 3: 3D Grid Cell Network with FCC lattice structure. (a) 3D activity packet representing position (x, y, z) through distributed population code. (b) FCC lattice structure showing hexagonal grid pattern with three height layers for dense 3D packing. (c) 4-DoF encoding scheme combining position and heading representations.

sian connectivity kernel. With a separable kernel

$$\mathcal{K}^{\mathrm{g}}_{a,b,c} = \mathcal{N}(a; 0, \sigma_x)\mathcal{N}(b; 0, \sigma_y)\mathcal{N}(c; 0, \sigma_z) \tag{40}$$

where $\mathcal{N}(\cdot; \mu, \sigma)$ denotes one-dimensional Gaussian distribution, and $\sigma_x, \sigma_y, \sigma_z$ are spatial kernel widths along three axes. the excitatory update is:

$$\Delta A^{\mathrm{g}}_{i,j,k} = \sum_{p,q,r} A^{\mathrm{g}}_{p,q,r}\, \mathcal{K}^{\mathrm{g}}_{\langle i-p\rangle_{N_x}, \langle j-q\rangle_{N_y}, \langle k-r\rangle_{N_z}} \tag{41}$$

where $\langle\cdot\rangle_N$ denotes modulo-N wrapping for periodic boundary conditions. Typical parameters are $\sigma_x = \sigma_y = \sigma_z = 5$ cells.

(2) Global inhibition and normalization. Uniform global inhibition subtracts the mean activity, while rectification and normalization maintain a single stable bump:

$$A^{\mathrm{g}}_{i,j,k} \leftarrow \left[A^{\mathrm{g}}_{i,j,k} + \Delta A^{\mathrm{g}}_{i,j,k} - \lambda\, \bar{A}^{\mathrm{g}}\right]_+, \quad \bar{A}^{\mathrm{g}} = \frac{1}{N_x N_y N_z} \sum_{p,q,r} A^{\mathrm{g}}_{p,q,r} \tag{42}$$

where $[\cdot]_+ = \max(0, \cdot)$ is rectification and λ controls inhibition strength (typically $\lambda = 1.0$). After inhibition, the activity is L_1-normalized:

$$A^{\mathrm{g}} \leftarrow \frac{A^{\mathrm{g}}}{\|A^{\mathrm{g}}\|_1} = \frac{A^{\mathrm{g}}}{\sum_{i,j,k} |A^{\mathrm{g}}_{i,j,k}|} \tag{43}$$

This normalization ensures the activity pattern has unit total mass, making the readout (center of mass) insensitive to overall activity magnitude.

(3) Path integration. The activity bump shifts according to fused velocity cues $\mathbf{v}^{\mathrm{f}}_t = (v^f_x, v^f_y, v^f_z)$. Index shifts are computed as:

$$(\Delta i, \Delta j, \Delta k) = \left(\lfloor \gamma_x v^f_x \Delta t \rfloor,\ \lfloor \gamma_y v^f_y \Delta t \rfloor,\ \lfloor \gamma_z v^f_z \Delta t \rfloor\right) \tag{44}$$

where $\gamma_x, \gamma_y, \gamma_z$ are metric-to-index gain factors converting physical velocity to network grid shifts.

$$A^{\mathrm{g}}_{i,j,k} \leftarrow A^{\mathrm{g}}_{\langle i-\Delta i\rangle_{N_x}, \langle j-\Delta j\rangle_{N_y}, \langle k-\Delta k\rangle_{N_z}} \tag{45}$$

This discrete shifting operation efficiently implements path integration on the toroidal network topology.

Visual Anchoring and Drift Correction. Pure path integration accumulates drift over time. To prevent unbounded error, the grid cell activity is anchored to visual landmarks through learned associations. Let $\Xi_{m,i,j,k}$ store the association strength between visual template m and grid cell (i, j, k). For each active visual template $m \in \mathcal{S}_{active}$, update associations and inject corrective activity:

$$\text{Hebbian learning: } \Xi^{t+1}_{m,i,j,k} = \max\left(\eta\, VT_m\, A^{\mathrm{g}}_{i,j,k},\ \Xi^{t}_{m,i,j,k}\right) \tag{46}$$

$$\text{Corrective injection: } \Delta A^{\mathrm{g}}_{i,j,k} \mathrel{+}= \frac{\rho}{|\mathcal{S}_{active}|} \sum_{m \in \mathcal{S}_{active}} VT_m\, \Xi_{m,i,j,k} \tag{47}$$

where η is Hebbian learning rate, ρ is spatial correction strength, and $\mathcal{S}_{active}$ is set of currently matched visual templates.

Multilayered Head Direction Cells. The multilayered head direction cell (HDC) network represents the robot's orientation using a 2D MD-CAN with dimensions $N_\psi \times N_h$ (azimuth bins by height layers). Different height layers allow representation of heading at various vertical positions, supporting 3D operation where orientation may vary with altitude.

Network Architecture. The multilayer head-direction network has activity $\mathbf{A}^{\mathrm{h}} \in \mathbb{R}^{N_\psi \times N_h}$ (typically $N_\psi = 360$ azimuth bins and $N_h = 5$ height layers). Each cell (u, h) is tuned to azimuth angle $\theta_u = 2\pi u / N_\psi$ and height layer h.

Attractor Dynamics. Similar to the grid cell network, the HDC network employs local excitation, global inhibition, and path integration. With a Gaussian kernel $\mathcal{K}^{\mathrm{h}}$:

$$\Delta A^{\mathrm{h}}_{\psi,h} = \sum_{u,v} A^{\mathrm{h}}_{u,v}\, \mathcal{K}^{\mathrm{h}}_{\langle \psi - u \rangle_{N_\psi}, \langle h - v \rangle_{N_h}} \tag{48}$$

followed by inhibition and normalization:

$$A^{\mathrm{h}} \leftarrow \frac{[A^{\mathrm{h}} + \Delta A^{\mathrm{h}} - \lambda_h\, \bar{A}^{\mathrm{h}}]_+}{\|A^{\mathrm{h}}\|_1} \tag{49}$$

Path integration shifts the activity bump according to yaw rate r^f and vertical velocity v_z^f:

$$\Delta\psi = \lfloor \gamma_\psi r^f \Delta t \rfloor, \quad \Delta h = \lfloor \gamma_h v_z^f \Delta t \rfloor \tag{50}$$

where γ_ψ converts angular velocity to azimuth bins, and γ_h maps vertical velocity to height layer shifts. **Pose Readout.** The metric pose estimate

is decoded from the spatial cell activities via center-of-mass computation (population vector decoding):

$$\hat{\mathbf{p}}_t = (\hat{x}_t, \hat{y}_t, \hat{z}_t) = \frac{\sum_{i,j,k} \mathbf{c}_{i,j,k} A^{\mathrm{g}}_{i,j,k,t}}{\sum_{i,j,k} A^{\mathrm{g}}_{i,j,k,t}}, \quad \mathbf{c}_{i,j,k} = (x_i, y_j, z_k) \tag{51}$$

$$\hat{\psi}_t = \operatorname{atan2}\left(\sum_{u,h} A^{\mathrm{h}}_{u,h,t} \sin\theta_u, \sum_{u,h} A^{\mathrm{h}}_{u,h,t} \cos\theta_u \right), \quad \theta_u = 2\pi u / N_\psi \tag{52}$$

This decoding naturally handles the circular topology of orientation (using atan2 to avoid angle wrapping issues) and provides sub-bin resolution for position through the weighted average.

3.5. Monocular Scale Recovery Branch

Problem. The 10th EKF state $\log s$ converges online from turn kinematics, which are observable only in turning frames (294/1101 frames on KITTI 07) and, by construction, recover only the *VO-relative* scale — the absolute scale between the visual frame and the metric world remains unrecoverable from inertial motion alone. Monocular metric-depth networks offer an independent cue, but their zero-shot predictions carry a scene-dependent bias (a preliminary study showed the raw depth scale to be unreliable across scenes), so raw depth cannot serve directly as a scale observation. Rather than trusting the depth network, we learn a lightweight *calibration* that maps depth-statistics features to the true global scale: trained once on simulation, frozen at deployment.

Structure. For each frame t, let $\hat{z}_t$ denote the metric depth map predicted by a frozen monocular depth foundation model (Depth-Anything-V2,

outdoor head). We summarize it into an 8-dimensional feature vector

$$\mathbf{f}_t = \left[\log \hat{s}_t^{\mathrm{dav2}},\ \log d_t^{\mathrm{med}},\ \log(d_t^{\mathrm{std}}+\epsilon),\ \log d_t^{\mathrm{p10}},\ \log d_t^{\mathrm{p90}},\ \log(\hat{s}_t^{\mathrm{inl}}+1),\ f_x/W,\ 1\right]^\top, \tag{53}$$

where $\hat{s}_t^{\mathrm{dav2}}$ is the windowed-median zero-shot depth scale estimate (VO displacement over median metric depth over the inter-frame interval), $d^{\mathrm{med}}, d^{\mathrm{p10}}, d^{\mathrm{p90}}, d^{\mathrm{std}}$ are the metric-depth median, percentiles, and standard deviation, $\hat{s}_t^{\mathrm{inl}}$ is the feature-inlier-consistent scale, and f_x/W encodes the camera geometry. A windowed median (W=50 intervals) smooths each feature. The scale head g_θ is a compact MLP 8→64→64→1 (ReLU, ∼4.8k parameters) predicting the log scale

$$\hat{\ell}_t = g_\theta(\mathbf{f}_t) \ \approx\ \log s_t^*, \tag{54}$$

supervised, during training only, by the per-interval ground-truth scale

$$\ell_t^* = \log \frac{\|\mathbf{p}_t^* - \mathbf{p}_{t-1}^*\|}{\|\mathbf{p}_t - \mathbf{p}_{t-1}\|}, \tag{55}$$

where $\mathbf{p}^*$ are ground-truth metric positions (available in simulation) and $\mathbf{p}$ the VO positions.

Train-Once Protocol. The head is optimized on three simulated towns (Town05/03/07) with Adam (learning rate 10^{-3}, weight decay 10^{-3}), MSE loss on standardized features, and early stopping on a held-out split (10%); the feature standardization is frozen together with the weights. A fourth town (Town10HD) is held out entirely, and no real-world data is seen. After training, *all parameters are frozen*: there is no per-sequence calibration and no online adaptation at deployment — the head is used as-is on every test sequence.

Gated Integration. The prediction $\hat{s}_t = \exp(g_\theta(\mathbf{f}_t))$ enters the EKF as the scale-branch observation in the log domain: it (i) replaces the depth-median initialization of $\log s$ when the branch activates, (ii) persistently pulls the scale state through a chi-square-gated update with a log-domain dead band, and (iii) suppresses the kinematic scale observation while active, so the two scale evidences do not fight. The correction is deliberately *scale-only*: it adjusts the 10th state and, through it, the visual-velocity observations — it never rewrites position or attitude directly, so a wrong scale prediction degrades gracefully rather than corrupting the fused trajectory.

Zero-Shot Behavior and Honest Limits. Within the simulation domain the learned mapping generalizes across scenes — including the held-out town — and outperforms an ordinary-least-squares calibration baseline on all five evaluation sequences. Across the sim-to-real boundary, however, the mapping does not transfer: on KITTI 07 the head predicts a scale correction in the wrong direction ($s \approx 0.3$ where the route requires ≈ 1.8), and deploying it degrades the EKF (Supplementary Section S.1 quantifies this failure). All real-world results in this paper therefore run without the scale branch — an honest limitation we return to in the experiments. The head's computational cost is negligible: $\sim$4.8k parameters and sub-millisecond CPU inference per frame, outside the 31-FPS critical path; the frozen depth backbone is evaluated asynchronously and cached.

3.6. Experience Map and Loop Closure

The multilayered experience map represents 3D spatial experiences as a graph structure, providing a hybrid topological-metric representation that combines the benefits of discrete place-based maps (topological robustness,

loop closure) and continuous metric representations (accurate localization).

Experience Representation. Each experience node $\mathcal{E}_i$ encodes a distinct spatial episode comprising the active visual template and descriptor (VT_i, $\boldsymbol{\phi}_i \in \mathbb{R}^{4096}$), the concurrent spatial-cell codes $\mathbf{P}_i^{gc} = \mathbf{A}_i^{\mathrm{g}} \in \mathbb{R}^{N_x \times N_y \times N_z}$ and $\mathbf{P}_i^{hdc} = \mathbf{A}_i^{\mathrm{h}} \in \mathbb{R}^{N_\psi \times N_h}$, the metric pose (x_i, y_i, z_i, ψ_i) decoded from these activities, and the timestamp t_i for temporal ordering. This rich multimodal encoding enables robust loop closure detection through visual matching while maintaining metric information through the spatial cell codes.

Experience Creation and Linking. As the robot navigates, new experiences are created when the current spatial representation differs sufficiently from all existing experiences. The creation criterion combines visual and spatial cell similarity:

$$\begin{aligned} \text{Create new experience if } \; & d_{cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_j) > \tau_{VT} \quad \text{for all } j \in \mathcal{E} \\ & \text{or } \|(\hat{x}_t, \hat{y}_t, \hat{z}_t) - (x_j, y_j, z_j)\| > d_{thresh} \end{aligned} \tag{56}$$

where $\tau_{VT} = 0.08$ is the visual template matching threshold and d_{thresh} is a spatial distance threshold (typically 0.5 m).

Consecutive experiences are linked by directed edges encoding the relative pose transition:

$$\mathbf{T}_{i \to i+1} = (\Delta x, \Delta y, \Delta z, \Delta\psi) = (x_{i+1} - x_i, y_{i+1} - y_i, z_{i+1} - z_i, \psi_{i+1} - \psi_i) \tag{57}$$

These edges form the primary topological structure of the experience map graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

Loop Closure Detection. Loop closure is detected when the current visual template matches a previous experience beyond a minimum temporal

separation (to avoid trivial closures):

$$\begin{aligned} \text{Loop closure detected if } \exists j : d_{cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_j) < \tau_{loop} \\ |t - t_j| > T_{min} \end{aligned} \tag{58}$$

where $\tau_{loop} = 0.05$ (stricter than the template creation threshold to reduce false positives) and $T_{min} = 50$ frames (approximately 5 seconds at 10 Hz).

Upon detecting a loop closure, a loop edge is added to the graph with the pose residual:

$$\mathbf{T}_{loop} = (\hat{x}_t - x_j, \hat{y}_t - y_j, \hat{z}_t - z_j, \hat{\psi}_t - \psi_j) \tag{59}$$

This loop constraint typically violates graph consistency (due to accumulated drift), triggering map relaxation.

Map Relaxation. Map relaxation distributes the loop closure error across all experiences along the loop path to minimize global inconsistency. We employ an iterative relaxation algorithm:

Step 1: Identify loop path. Given a loop closure edge from experience j to experience t, identify the sequence of experiences $\mathcal{P} = \{j, j+1, \ldots, t-1, t\}$ along the path.

Step 2: Compute pose residual. The accumulated pose transition along the loop should ideally close (return to the origin):

$$\mathbf{e}_{loop} = \sum_{k \in \mathcal{P}} \mathbf{T}_{k \to k+1} - \mathbf{T}_{loop} \tag{60}$$

where $\mathbf{e}_{loop}$ is accumulated pose drift residual around closed trajectory, and $\mathcal{P}$ denotes ordered experience sequence forming the loop. **Step 3: Distribute correction.** Distribute the correction proportionally across the loop experiences:

$$\mathbf{T}_{k \to k+1} \leftarrow \mathbf{T}_{k \to k+1} - \frac{\mathbf{e}_{loop}}{|\mathcal{P}|} \tag{61}$$

This uniform distribution assumes drift accumulates at a constant rate, a reasonable first-order approximation.

Step 4: Update experience poses. Recompute all experience poses by integrating the corrected transitions:

$$(x_{k+1}, y_{k+1}, z_{k+1}, \psi_{k+1}) = (x_k, y_k, z_k, \psi_k) + \mathbf{T}_{k \to k+1} \tag{62}$$

Step 5: Re-anchor spatial cells. Inject the corrected pose $(\hat{x}, \hat{y}, \hat{z}, \hat{\psi})_{corr}$ back into the spatial cell networks by reinitializing the activity bumps at the corrected locations. This "resets" the path integration from the corrected reference, preventing error accumulation in subsequent operation.

This relaxation procedure runs iteratively until convergence or a maximum number of iterations, ensuring the experience map remains globally consistent while preserving local metric accuracy.

Deployed Loop-Closure Line: Frame-Based Candidates. The reported system instantiates loop-closure detection over *frames* rather than experience nodes: every frame is described by a frozen DINOv2-S global scene embedding $\mathbf{e}_t$, refined by a fine-tuned head trained under the same train-once protocol as the scale branch (simulation data only). A candidate match for the current frame is the historical frame $j^* = \arg\max_j \cos(\mathbf{e}_t, \mathbf{e}_j)$ restricted to frames whose EKF-odometry arc-length separation exceeds $\text{arc}_{\min}$ (50 m), which removes trivial self-matches while keeping the search exhaustive over the plausible far-history. This frame-level candidate stage decouples loop detection from experience-node creation and provides dense per-frame evidence to the gate chain.

Four-Gate Chain. A candidate becomes an accepted loop event only if

it passes four fixed gates (all thresholds set once across all five scenarios):

$$\text{accept}(t, j^*) \iff \cos(\mathbf{e}_t, \mathbf{e}_{j^*}) \geq \tau \wedge \text{streak}(t, j^*) \geq 15 \wedge |\Delta\psi_{tj^*}| \leq 60^\circ \wedge \|\mathbf{c}_{tj^*}\| \leq 25\,\text{m}, \tag{63}$$

with a cumulative correction budget $\sum_i \|\mathbf{c}_i\| \leq 35$ m across the trajectory and deduplication of historical anchors (each past frame anchors at most once). The *similarity gate* ($\tau = 0.94$) is calibrated by a five-scenario sweep: observed false candidates score 0.907–0.935 while strong genuine revisits score ≥ 0.949. The *persistence gate* (≥ 15 consecutive candidate frames, ± 10 tolerance) exploits a structural property of genuine revisits: a vehicle traversing the same street produces a sustained streak of high-similarity frames, whereas perceptual aliasing produces single-frame or short-burst excursions — this gate is scale-free and transfers across domains unchanged. The *orientation gate* (60° heading difference) admits same-direction and reversed re-visits while rejecting geometrically implausible matches. The correction budget bounds the damage of any single wrong acceptance. The chain is deliberately conservative: on KITTI 09 its sole genuine loop lies in the final two frames, where the streak can reach only $2 < 15$ and the gate correctly declines — releasing it in hindsight would change the error by merely -0.12 m — while the orientation gate rejects that sequence's two high-similarity but geometrically inconsistent candidates (Supplementary Section S.3). Prefer silence to a wrong re-anchor.

Injection. An accepted event injects a one-shot translation correction $\mathbf{c}_{tj^*}$ (step size $\beta = 1.0$) that pulls the current segment toward the matched node's map pose, writes a link edge into the experience graph, and hands the event to the full-Sim(3) re-anchoring operator described below; the corrected segment then enters the drift-field relaxation above.

Distributed Closure Drift Field. Each re-anchoring event (a loop closure or a match to an existing node) resets the incremental dead-reckoning state, so the raw output trajectory contains a discrete correction step of $\mathbf{c}_i$ at frame f_i. These steps form an accumulated staircase $\mathbf{C}_{\text{cum}}(f)$ whose discrete jumps are the dominant source of per-frame inconsistency between the experience-map-corrected output and the filtering front-end. We therefore relax each step into a continuous drift field, inspired by slow re-consolidation of spatial memory in the hippocampal–entorhinal system. For every recorded re-anchoring pulse $\mathbf{c}_i$ at frame f_i with $\|\mathbf{c}_i\| \le C_{\max}$, a raised-cosine ramp of duration W_i is applied:

$$F_i(f) = \begin{cases} \frac{1}{2}\big(1 - \cos(\pi(f - f_i)/W_i)\big), & f_i \le f < f_i + W_i \\ 1, & f \ge f_i + W_i \end{cases} \tag{64}$$

and the corrected trajectory is

$$\hat{\mathbf{p}}(f) = \mathbf{p}_{\text{exp}}(f) + \Big(\sum_i F_i(f)\,\mathbf{c}_i \;-\; \sum_{i:\, f_i \le f} \mathbf{c}_i\Big), \tag{65}$$

so each correction persists (the held level $\mathbf{c}_i$ is retained after the ramp) but its onset is smoothed over W_i frames, and the staircase $\mathbf{C}_{\text{cum}}$ is exactly replaced by the continuous field. Three design choices, validated by offline parameter sweeps on all five outdoor sequences, shape the field:

- **Adaptive relaxation time.** $W_i = \text{clip}(W_0 + W_\perp \|\mathbf{c}_i\|,\ W_0,\ W_{\max})$: larger jumps relax more slowly, analogous to the slower re-consolidation of stronger spatial memories and to the longer time constants of grid-cell drift correction.

- **Tail completion.** Each ramp is truncated at the trajectory end ($W_i \leftarrow \min(W_i, f_{\text{end}} - f_i + 1)$) so that the drift field is fully relaxed by the final frame, avoiding a half-relaxed ramp that would leave terminal bias.

- **Amplitude gating.** Jumps with $\|\mathbf{c}_i\| > C_{\max}$ are applied instantaneously and excluded from the field: large corrections correspond to distant true loop closures whose early node positions are trustworthy, whereas small corrections (nearby re-visits, hash collisions) are uncertain and benefit from slow relaxation. Sweeps show that flattening *all* steps degrades accuracy (e.g., Town01: $+0.64\%$ RMSE), while gating yields per-frame RMSE improvements over both the un-relaxed baseline and the EKF front-end on all five sequences.

With the EKF-odometry mode (NLM_USE_EKF_ODO) the experience map consumes the fused EKF trajectory as its motion source (frame-to-frame displacement and yaw increments), and new experience nodes are pinned to the EKF map coordinates, so the grid-cell path integrator and the experience graph operate on a common, tightly constrained metric ground truth; the drift field then corrects only the residual re-anchoring steps on top of this unified foundation.

Closed-Loop Full-Sim(3) Re-Anchoring. Each re-anchoring pulse above translates the subsequent trajectory but does not change its global scale or orientation, so the monocular scale ambiguity keeps accumulating between loops. We therefore complement it with a loop-triggered operator: at every accepted loop-closure event at frame t_{now}, a full similarity transform is fitted between the *anchor window* $\mathbf{A}$ (the first win moving frames of the trajectory, after the static prelude) and the *current window* $\mathbf{C}$ (the win frames

ending at t_{now}, capped by the next event) by a Procrustes fit

$$\mathbf{A} \approx s\,(\mathbf{CR}) + \mathbf{t}, \qquad s \in [0.05,\, 20], \tag{66}$$

with win = 250 frames (self-calibrated by an offline window sweep) and the scale gate rejecting degenerate fits. The entire segment after t_{now} (up to the next event) is then pulled back into the anchor frame,

$$\hat{\mathbf{p}}(f) = s\,(\mathbf{p}(f)\mathbf{R}) + \mathbf{t}, \qquad t_{\text{now}} < f < t_{\text{next}}, \tag{67}$$

applied independently per event on the pre-correction trajectory (non-chained, so one event's refit is not polluted by its predecessor's transform). Because each accepted loop is a genuine re-visit of the same physical location, the re-anchoring is a global consistency correction that uses no ground-truth information; on loop-free routes it is a strict no-op. On a dedicated closed-loop benchmark (an urban route crossing the same road segments at widely separated times), this full-Sim(3) re-anchoring recovers 9× more of the accumulated global drift than the EKF baseline (44.24 m vs 398.05 m, anchored-Sim(3) caliber): the dominant global error accumulated between loops is the monocular scale error, which only the full similarity re-fit removes.

4. Experiments and Results

We evaluate NeuroLocMap on five scenarios spanning simulated urban driving (CARLA Town05/03/07/10HD) and real-world highway driving (KITTI 07). The full system couples the vestibular-visual EKF core with two higher-level corrections: an online monocular scale branch (a lightweight regression head over a frozen depth foundation model, trained once on simulated data

and frozen at deployment) and an experience-map loop-closure line built on learned scene embeddings. All higher-level components are *trained once on simulation* and deployed *zero-shot* on every test sequence, including the real-world one.

4.1. Experimental Setup and Datasets

Implementation Details. All experiments are conducted on a laptop workstation equipped with an Intel Core i5-12450H CPU (12th Gen, 8 cores), 16 GB RAM, and NVIDIA GeForce RTX 4050 Laptop GPU. The visual processing pipeline leverages GPU acceleration for HART-like feature extraction and Transformer-inspired encoding (using simplified self-attention via global statistical modulation), while spatial cell computations run on CPU. The LSTM temporal module uses manually-tuned fixed gate values (input=0.55, forget=0.6, output=0.92) rather than learned parameters. The framework achieves real-time performance at $\sim$31 FPS on average across all test scenarios.

Baseline Methods. We compare NeuroLocMap against two baseline approaches: (1) EKF Fusion — a loosely-coupled 10-state Extended Kalman Filter $(x, y, z, v_x, v_y, v_z, \phi, \theta, \psi, \log s)^\top$, where the 10th state $\log s$ is the online-estimated logarithm of the monocular VO scale (initialized to 0 and converging online from inertial observability). The state is propagated with the IMU (gyroscope attitude integration plus bias-compensated accelerometer dead reckoning, with a static-startup bias estimate) and corrected with monocular visual-odometry measurements through four gated observation branches: (i) a 1-D horizontal velocity-magnitude observation (VO inter-frame displacement $\times$ online scale, over the true inter-update interval of IMU step count

$\times\Delta t$) that drives velocity correction and, when the innovation exceeds the gate, directly reinitializes the velocity state; (ii) a roll/pitch attitude observation; (iii) a short-window relative position observation re-anchored to the current filter position every 30 frames, built from the gyro-integrated heading direction $\times$ VO displacement magnitude $\times$ online scale, so that the global VO heading drift does not enter the fused position; and (iv) an independent scale observation estimating $\log s$ from turn kinematics (lateral specific force over yaw rate, $|a_{\text{lat}}|/|\omega|$, normalized by the recorded VO speed, windowed median). Each branch is guarded by a per-branch Mahalanobis chi-square gate (0.99–0.999 level) to reject outliers. The observation noise matrix R is residual-adaptive (sliding-window smoothed innovation mapped to $[R_{\min}, R_{\max}]$, combined with a VO inlier-count quality factor), and the covariance is updated in Joseph form with post-update regularization (symmetrization + minimum-eigenvalue clamping). This is a representative classic visual–inertial filter estimator in the family of MSCKF/VINS-Mono [15, 16]; and (2) Visual Odometry (VO) — a feature-based monocular VO baseline using ORB features with epipolar-geometry pose recovery (Essential matrix + `recoverPose`), without IMU fusion or loop closure. We additionally report *EKF+scale*: the same 10-state EKF whose scale state is driven by the deep scale head of the Methods section (no loop closure), which isolates the scale-branch contribution; and, where meaningful, the *loop-closure line alone* (the experience-map pipeline without the scale branch).

Implementation-Level Fairness. For all methods, we use the same synchronized sensor streams and timestamps. Baseline trajectories are generated by our Python pipeline: on CARLA they are recorded during data

collection (the EKF fusion baseline to fusion_pose.txt, the VO baseline to visual_odometry.txt, and the ground truth to ground_truth.txt); on KITTI 07 the same EKF/VO pipeline is run offline on the raw public sequence with identical configuration. The evaluation then loads these trajectories and compares them against NeuroLocMap under identical sequences, ensuring differences reflect algorithmic design rather than data or timing mismatches. All reported CARLA numbers correspond to data collected with the final EKF implementation (single global hyper-parameter set shared by all maps; collection parameters: agent speed 12 km/h, safe following distance 6 m, background traffic of 20 autopilot vehicles and 20 kinematically driven pedestrians), so the numbers in Table 3 are directly reproducible from the public repository.

Evaluation Datasets. We evaluate on five diverse scenarios across two benchmark datasets (Table 2): (1) CARLA Simulation (Town05/03/07/10HD): simulated urban driving scenarios [34] with ground-truth poses and synchronized IMU-camera data, all re-collected with the same protocol and background traffic. Town05 (486 m) is the loop-richest route, crossing the same commercial streets repeatedly; Town03 (474 m) threads narrow alleys with strong viewpoint changes; Town07 (618 m) is a low-texture suburban loop; Town10HD (452 m) is an open layout with essentially a single revisit. (2) KITTI Odometry Sequence 07 [35]: real-world highway driving (692 m, 1101 frames) with high-quality visual features and limited loop opportunities, challenging systems on medium-length real-world trajectories.

Evaluation Metrics and Alignment Calibers. We employ standard SLAM evaluation metrics: Absolute Trajectory Error (ATE, RMSE after

Table 2: Dataset Characteristics and Evaluation Scenarios. All CARLA scenarios were re-collected at 12 km/h with the final EKF implementation and background traffic (20 autopilot vehicles + 20 kinematically driven pedestrians); KITTI is a fixed public benchmark. "Accepted links" counts the loop-closure edges admitted by the persistence and orientation gates.

Dataset	Environment	Length	Frames	Characteristics
Town05	Urban mixed	486 m	5000	Loop-rich route, 7 accepted links
Town03	Urban alleys	474 m	5000	Narrow streets, viewpoint changes, 7 links
Town07	Suburban loop	618 m	5000	Low-texture residential loop, 7 links
Town10HD	Urban open	452 m	5000	Open layout, single revisit, 1 link
KITTI 07	Highway	692 m	1101	Real-world open trajectory, 11 map links

alignment) [36], terminal drift rate (terminal error normalized by trajectory length), and Relative Pose Error (RPE, frame-to-frame consistency). Because NeuroLocMap recovers the metric scale online, the alignment protocol is itself part of the claim, and we evaluate *every* method under two calibers, each estimated over the full trajectory and applied consistently to all methods: (i) *full Sim(3) refit* (rotation, translation, and a single global scale, Umeyama/Procrustes) — by construction the fitted scale absorbs any global scale error, so this caliber measures trajectory *shape* and serves as an upper bound for all methods; and (ii) *SE(3) rigid alignment* (Kabsch, rotation and translation only, no scale degree of freedom) — a trajectory whose metric scale is wrong cannot be rescued by the aligner, so this caliber directly tests metric-scale recovery. Our metric-localization claim is stated in the SE(3) caliber: NeuroLocMap outperforms the EKF without any post-hoc scale fitting. We do not claim a globally scale-perfect system: residual global scale

errors remain (Sim(3)-fitted scale factors 0.79–1.92 across scenarios), and the Sim(3) numbers should be read as the shape-only upper bound.

4.2. Performance Results and Analysis

Quantitative Performance. Table 3 presents the quantitative comparison under both calibers. NeuroLocMap (full system) *outperforms the 10-state EKF visual-inertial baseline on all five scenarios under both calibers*: by +16.5% (KITTI 07) to +59.3% (Town05) under the Sim(3) upper bound, and by +1.5% (KITTI 07, the thinnest margin) to +33.9% (Town05) under the SE(3) rigid caliber — where an unrecovered metric scale cannot be absorbed by the aligner. The *scale branch is the dominant contributor*: driving the EKF's scale state with the deep scale head alone (EKF+scale, no loop closure) already removes 13.1–32.7% of the EKF's SE(3) error and 22.4–54.8% of its Sim(3) error, and the full system adds a further loop-closure margin on Town05 (−2.03 m Sim(3)) and Town03 (−0.76 m). Against monocular VO, the full system wins on all five scenarios by +33.7% to +68.8% (Sim(3)). Two honest caveats accompany these headline numbers. First, on Town07 the scale branch *alone* is the strongest configuration (27.90/29.21 m vs 28.75/31.51 m for the full system): the loop's marginal contribution there is slightly negative (+0.85 m), which we analyze in Section 4.5. Second, on KITTI 07 the scale head does not transfer zero-shot to real imagery (Supplementary Section S.1), so no EKF+scale row exists there and the KITTI result is carried by the loop-closure line alone.

Performance by Scenario Type. On the four CARLA towns (452–618 m), NeuroLocMap cuts the EKF's error by 22.4% (Town10HD) to 59.3% (Town05) under Sim(3), and by 13.1% to 33.9% under SE(3), while cutting

Table 3: Quantitative Performance Comparison Under Two Alignment Calibers (ATE RMSE, m)

	Sim(3) full refit (shape-only upper bound)				SE(3) rigid (metric-scale caliber)			
Dataset	NeuroLocMap	EKF+scale	EKF	VO	NeuroLocMap	EKF+scale	EKF	VO
Town05	**18.07**	20.10	44.44	48.26	**32.67**	33.26	49.40	53.86
Town03	**15.31**	16.07	23.36	49.10	**17.31**	17.47	23.53	49.15
Town07	28.75	**27.90**	40.64	50.07	31.51	**29.21**	42.85	50.13
Town10HD	**14.44**	**14.44**	18.61	35.86	**49.88**	49.89	57.38	58.04
KITTI 07	**18.45**	n/a	22.09	27.84	**29.99**	n/a	30.45	38.35

Bold = best of the four configurations in that caliber. NeuroLocMap beats the EKF on all five scenarios under both calibers (Sim(3) +16.5% to +59.3%; SE(3) +1.5% to +33.9%) and monocular VO on all five (Sim(3) +33.7% to +68.8%).

Drift rates (terminal error / length): NeuroLocMap 2.9–11.7% (Sim(3)) and 0.2–15.2% (SE(3)), versus 1.6–13.3% and 5.0–19.0% for the EKF and 9.9–24.2% and 13.7–17.2% for VO.

n/a: the zero-shot scale head is not deployed on KITTI 07 (real-image domain gap; Supplementary Section S.1); the KITTI result is the loop-closure line over the EKF core. On Town07 the scale branch alone is strongest (loop contribution +0.85 m); elsewhere the full system is best.

monocular VO's error by 42.6–68.8%. The gains decompose cleanly: the scale branch supplies the bulk (EKF→EKF+scale), and the experience map contributes a further 0–2.0 m where the route is genuinely revisited (7 accepted links on Town05/03/07 versus a single inert link on Town10HD). The urban experience maps hold 91–481 experience nodes (Town05: 175, Town03: 91, Town07: 270, Town10HD: 481) with 91–481 visual templates.

On the real-world 692 m highway (KITTI 07), NeuroLocMap runs with the loop-closure line over the EKF core (the zero-shot scale branch is disabled; Supplementary Section S.1) and still outperforms the EKF by 16.5% under Sim(3) (18.45 m vs 22.09 m) and by 1.5% under SE(3) (29.99 m vs 30.45 m),

while beating monocular VO by 33.7% (27.84 m). This validates that the vestibular-visual core plus learned loop closure remains effective on real-world driving — and quantifies, honestly, how much of the framework's simulated-data advantage currently survives the domain gap.

Representative Performance Analysis. Figure 4 visualizes two representative scenarios: Town05 (486 m, loop-rich mixed urban) and Town10HD (452 m, open layout). The top row shows per-method metrics (RMSE, drift, RPE, and template/loop counts); the bottom row shows the aligned per-frame position error evolution. On Town05, NeuroLocMap reaches 18.07 m RMSE — 59.3% better than the EKF (44.44 m) and 62.6% better than monocular VO (48.26 m, whose drift reaches 24.2% versus 4.6%) — while its experience map holds 175 nodes over 175 visual templates with 7 accepted links. On Town10HD, it reaches 14.44 m versus 18.61 m for the EKF (+22.4%) and 35.86 m for VO (+59.7%); the map holds 481 nodes but only a single accepted link, and the EKF+scale row (14.44 m) shows that essentially the entire gain comes from the scale branch, the loop being inert on this open layout. These two cases bracket the framework's operating regime: loop-rich urban driving, where experience-based re-anchoring adds global consistency on top of the metric-scale correction, and open layouts, where the system reduces to EKF+scale. Comprehensive results for all five datasets are given in Table 3.

Framework Internal Representations. The topological structure of the experience maps varies with scene revisiting structure: the CARLA maps hold 91–481 experience nodes (Town05: 175, Town03: 91, Town07: 270, Town10HD: 481) with 91–481 visual templates and 1–7 accepted PR links,

while the KITTI 07 highway map is far denser (964 nodes, 964 templates over 692 m, $\sim$1.4 templates per meter), reflecting the high sampling rate of a visually distinctive, non-repeating scene. On loop-rich urban maps, the accepted re-anchors provide the global constraints that relax accumulated drift between revisits. The bio-inspired spatial cell representations demonstrate characteristic periodic firing patterns at different spatial scales (1.5 m to 12 m wavelength) for grid cells, and orientation-selective neural activity for head direction cells, confirming successful integration of vestibular (IMU) signals for orientation estimation.

Computational Performance and Complexity Analysis. Table 4 details per-module timing analysis. NeuroLocMap achieves real-time performance at $\sim$31 FPS average, with total pipeline latency of 32.1 ms. The grid cell update dominates computation (58% of total time) due to 3D continuous attractor network dynamics, while experience map update (15%) and visual template matching (7%) remain lightweight. Visual processing achieves high throughput (2.1 ms, 469 Hz) despite dual-stream architecture, benefiting from the simplified $\mathcal{O}(N)$ self-attention implementation (global statistical modulation rather than full QKV operations) and efficient feature caching. The deep components add negligible cost: the frozen depth backbone and the scale head run outside the 31-FPS critical path (the head itself is $<$1 ms per frame on CPU), and the loop-closure line consumes precomputed frame embeddings.

Theoretical Complexity. The framework's computational complexity scales as: (1) Visual processing: $\mathcal{O}(HW)$ for image preprocessing plus $\mathcal{O}(N_f)$ for feature extraction, where $H \times W = 640 \times 480$ and $N_f \approx 300$ features; (2)

Grid cell update: $\mathcal{O}(N_x N_y N_z)$ for 3D attractor dynamics with $30 \times 30 \times 10 = 9000$ cells; (3) Template matching: $\mathcal{O}(M \cdot d)$ where M is template count (91–964) and d is descriptor dimension (4096); (4) Experience map: $\mathcal{O}(K)$ for graph traversal with K nodes (91–964). Total complexity is $\mathcal{O}(HW + N_x N_y N_z + Md)$, dominated by grid cell dynamics. Compared to learning-based methods requiring GPU matrix operations ($\mathcal{O}(N^2)$ for attention), our $\mathcal{O}(N)$ approximation enables CPU-only real-time execution.

Hardware Requirements and Deployment. The framework runs on commodity hardware without GPU acceleration for the SLAM pipeline itself: Intel Core i5-12450H CPU (8 cores, 2.0–4.4 GHz), 16 GB RAM, consuming ∼2 GB memory, dominated by the runtime state of the 3D grid-cell attractor network (visual template storage is ∼16 MB even in the densest map: 964 templates × 4096-dim descriptors × 4 bytes, plus experience map overhead). Peak CPU utilization reaches 45–60% during grid cell updates, leaving headroom for concurrent processes. This lightweight footprint enables deployment on embedded platforms (e.g., NVIDIA Jetson Nano, Raspberry Pi 4) where learning-based SLAM methods requiring GPU inference (DROID-SLAM: 8 GB VRAM, TartanVO: 6 GB VRAM) are infeasible. Power consumption is estimated at 15–20 W (CPU-only) vs 50–80 W for GPU-accelerated methods, critical for battery-powered mobile robots.

4.3. Component Ablation

Scale Branch vs. Loop Line. Figure 5(a) decomposes the full system on Town05 (486 m, Sim(3) caliber): the full system reaches 18.07 m; removing the loop line (EKF+scale, no closure) yields 20.10 m; removing the scale branch (loop line over the unscaled EKF core) yields 41.14 m; the bare EKF

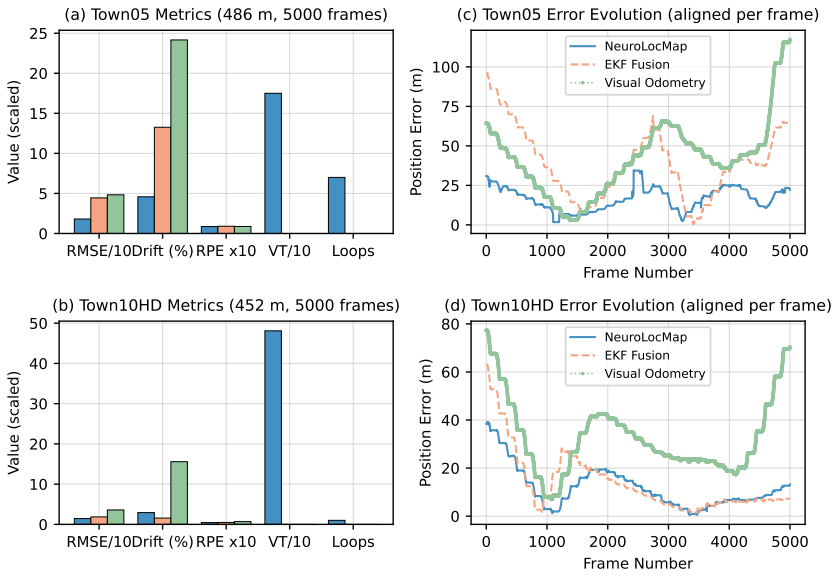


Figure 4: Representative performance for Town05 (486 m, loop-rich) and Town10HD (452 m, open layout). Top row: per-method metrics (RMSE, drift, RPE, template/loop counts); bottom row: aligned per-frame position error. On Town05 NeuroLocMap (18.07 m, full system) is 59.3% better than the EKF (44.44 m) and 62.6% better than VO (48.26 m); on Town10HD it reaches 14.44 m vs 18.61 m for the EKF (+22.4%), with the EKF+scale row showing the gain there is carried entirely by the scale branch (single inert loop link).

is 44.44 m and VO 48.26 m. Two conclusions follow. First, the *scale branch is the single dominant component*: it removes 24.3 m of the total 26.4 m gap between the EKF and the full system, consistent with the global scale being the largest monocular error source on this route. Second, the loop line contributes a further −2.03 m on top of the scaled core — a real but secondary margin that scales with genuine revisiting structure (Section 4.5 quantifies where it vanishes). The same decomposition holds on Town03 (full 15.31 / EKF+scale 16.07 / loop-only 22.91 / EKF 23.36 m) and Town10HD,

Table 4: Runtime Performance Analysis (Averaged Across the Four CARLA Towns)

Component	Mean (ms)	Frequency (Hz)
Visual processing (dual-stream)	2.1	469
Visual template matching	2.3	429
Vestibular-visual fusion	1.9	527
Grid cell update (3D CAN)	18.5	54
HDC update	1.5	661
Experience map update	4.7	213
Total pipeline	**32.1**	**31.2**

where the loop is inert and full system $\equiv$ EKF+scale (14.44 m).

Learned Embeddings vs. Classical Descriptors. Within the loop-closure line, replacing the learned scene embeddings with classical ORB-bag descriptors collapses the loop benefit: on Town05 the ORB-descriptor loop line yields 44.72 m — statistically indistinguishable from the bare EKF (44.44 m) — whereas the learned-embedding loop line reaches 41.14 m (Sim(3)). The persistence and orientation gates do the same work in both cases; it is the embedding quality that decides whether a revisited street is recognized at all.

Visual Template Economy. Figure 5(b) shows visual template accumulation across the five scenarios: the framework grows 91–964 templates depending on scene repetitiveness, versus the $\sim$5 templates typical of classical RatSLAM [37]. The densest map is the visually non-repeating KITTI 07 highway (964 templates over 692 m, $\sim$1.4 per meter); the CARLA maps hold 91–481 (Town05: 175, Town10HD: 481, Town07: 270, Town03: 91), reflecting how much of each route is genuinely re-observed. Growth is monotone

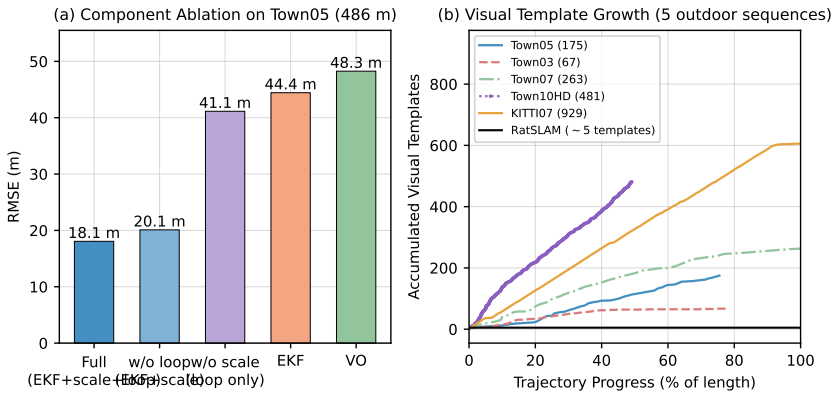


Figure 5: Component ablation. (a) Town05 (486 m) component decomposition, Sim(3) caliber: full system (EKF+scale+loop, 18.07 m) vs. w/o loop (EKF+scale, 20.10 m), w/o scale (loop line, 41.14 m), EKF (44.44 m), VO (48.26 m). (b) Visual template growth vs. trajectory progress for the five evaluation sequences. The black line marks the RatSLAM baseline (∼5 templates [37]).

(templates are never deleted), and this selective, scene-adaptive template economy enables fine-grained place discrimination without unbounded map growth.

4.4. Discussion and Insights

Performance Summary. Figure 6 summarizes the five-scenario comparison. Panel (a) shows NeuroLocMap below both baselines everywhere except the Town07 Sim(3) column, where EKF+scale is marginally best; panel (b) shows the improvement over the EKF under Sim(3) (+16.5% to +59.3%, bubble size proportional to route length); panel (c) shows the improvement over monocular VO (+33.7% to +68.8%). With only five scenarios we refrain from correlation statistics; the qualitative pattern is that the *scale branch* contribution tracks the severity of the monocular scale error of each route

(largest on Town05 and Town10HD, where the Sim(3)-fitted scale factors are 1.51 and 1.92), while the *loop-line* contribution tracks genuine revisiting structure (7 accepted links on Town05/03/07, one inert link on Town10HD).

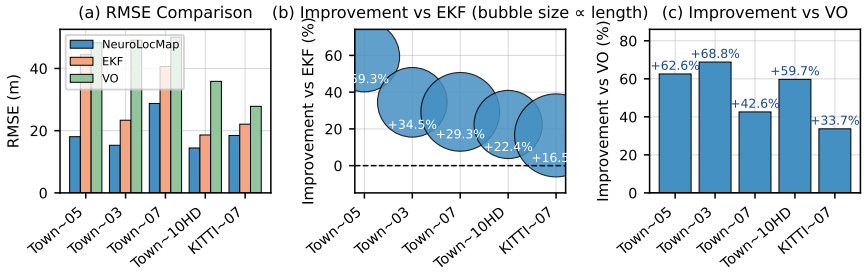


Figure 6: Performance summary across the five evaluation scenarios. (a) RMSE comparison for NeuroLocMap (full system), EKF+scale, EKF, and VO (Sim(3) caliber). (b) Improvement vs. EKF (bubble size indicates trajectory length): +16.5% to +59.3%. (c) Improvement vs. monocular VO: +33.7% to +68.8%.

When Does NeuroLocMap Excel? Three conditions emerge:

- **Monocular Scale-Error Regime** — the scale branch pays off wherever the VO scale error is large and systematic: on Town10HD the raw EKF converges to a fitted scale of 0.65 (a 35% scale error) yet 18.61 m Sim(3) / 57.38 m SE(3); with the scale head the same core reaches 14.44 m / 49.88 m. The SE(3) caliber is where this matters: it is exactly the error the Sim(3) refit would otherwise absorb.

- **Revisit-Rich Layouts** — the experience map converts revisits into global constraints: Town05 (7 links) gains a further −2.03 m on the scaled core, Town03 (7 links) −0.76 m, whereas Town10HD's single inert link contributes nothing — the system degrades gracefully to EKF+scale rather than being hurt.

- **Real-World Transfer with Honest Limits** — on KITTI 07 the loop-closure line transfers zero-shot and delivers the +16.5% (Sim(3)) / +1.5% (SE(3)) margin; the scale head does not (Supplementary Section S.1). The framework therefore remains effective on real data, but its simulated-data scale advantage is currently sim-only, and we say so.

4.5. Failure Case Analysis

Town07: Loop Contribution Slightly Negative. Town07 is the one scenario where the full system (28.75 m Sim(3) / 31.51 m SE(3)) is marginally *worse* than EKF+scale (27.90/29.21 m): its single accepted-loop event contributes +0.85 m. The route is a low-texture suburban loop whose revisits are visually ambiguous; the embedding similarity of the accepted event sits close to the $\tau = 0.94$ gate, and the resulting re-anchoring pulls a segment toward a slightly wrong map pose. The persistence gate (15 frames) and orientation gate (60°) filter most such events elsewhere, but Town07 shows the residual false-positive risk of embedding-only loop verification. A geometric consistency check on accepted loops is the natural remedy and is left for future work.

KITTI 07: Zero-Shot Scale-Head Failure. The scale head, trained once on CARLA, predicts scale corrections in the wrong direction on real KITTI imagery (predicting $s \approx 0.3$ where the route requires ≈ 1.8; Supplementary Section S.1). Deploying it on KITTI degrades the EKF from 17.8 m to 23.1 m (Sim(3), real-VIO baseline) in our ablation, so the full-system KITTI result deliberately runs without it. This is the clearest evidence in the paper of the sim-to-real gap for the learned scale branch — the loop line, whose embeddings transfer, carries the KITTI result instead.

Town10HD: Inert Loop. With a single accepted link that never fires, the full system equals EKF+scale bit-for-bit (14.44 m). This is by design (the gates prefer silence to a wrong re-anchor), and it bounds the downside of the loop machinery on open layouts.

Summary. NeuroLocMap excels on routes with large monocular scale error and genuine revisiting structure, from the 452–618 m CARLA towns to the 692 m real-world highway. Its two learned components degrade differently under domain shift: the loop embeddings transfer zero-shot, the scale head currently does not.

4.6. Input-Output Visualization

Figure 7 visualizes the framework's input-output relationship on the KITTI 07 highway sequence (1101 frames, 692 m). Panel (a) shows representative input images with heading indicators alongside IMU data; panel (b) shows the trajectory comparison — NeuroLocMap 18.45 m RMSE vs. EKF 22.09 m and VO 27.84 m (Sim(3)) — and the experience map topology: 964 nodes (964 visual templates) connected by sequential links and 11 loop/reanchor edges detected through hippocampal-like place recognition. On this real-world sequence the framework transforms multimodal sensory inputs into coherent spatial representations for both localization and mapping, with the loop line providing the global constraints that the EKF core lacks.

Comparison with Classical VIO/SLAM Baselines. To situate NeuroLocMap beyond the internal EKF/VO baselines, Table 5 compares it with published visual, visual-inertial, and SLAM systems using the official KITTI odometry metric, the relative trajectory error (RTE): the mean per-frame translation drift over 100–800 m subsequences, in % of traversed distance

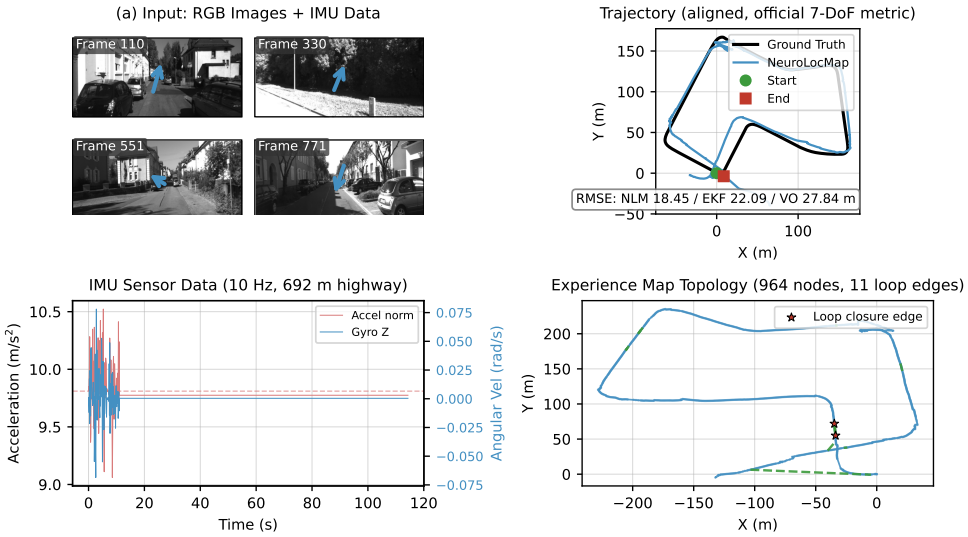


Figure 7: Input-output visualization for the KITTI 07 highway sequence (692 m, 1101 frames). (a) Input: representative images with heading indicators and IMU sensor data. (b) Output: trajectory comparison (ground truth vs. NeuroLocMap, RMSE 18.45 m vs. EKF 22.09 m, Sim(3)) and the experience map topology with 964 nodes and 11 detected loop/reanchor edges.

Table 5: Comparison with published baselines on the KITTI odometry benchmark. RTE: relative trajectory error (% of traversed distance); ATE: 7-DoF Sim(3)-aligned RMSE on sequence 07 (m).

Method	Sensors	RTE (%)	ATE (m)
SOFT2 [38]	stereo	0.53	—
SOFT-SLAM [39]	stereo	0.65	—
VINS-Fusion [40]	stereo+IMU	1.09	—
ORB-SLAM2 [41]	stereo	1.15	—
MonoDepth2 [42]	monocular	12.59	—
NeuroLocMap (EKF core)	mono+IMU	94.7 / 72.1	22.09
NeuroLocMap (full)	mono+IMU	101.1 / 75.2	**18.45**

Published RTE: KITTI odometry benchmark values as reported by each system; stereo systems are intrinsically scale-consistent, and MonoDepth2 (a learned depth network with metric output) is the only purely monocular entry.
SOFT2 is the current top entry on the official KITTI odometry leaderboard; its 0.53% submission uses recalibrated camera parameters.
NeuroLocMap RTE on sequence 07: raw (unaligned) / after 7-DoF Sim(3) alignment. ATE: Sim(3)-aligned RMSE.

(a per-frame odometry metric, distinct from the terminal/length drift rate reported in Table 3). Published values are the KITTI odometry benchmark figures reported by each system; our values are measured on sequence 07 under two protocols.

Two honest observations follow. First, every scale-consistent published system sits below 1.2% RTE: stereo geometry fixes the scale intrinsically, and the sole monocular entry is a learned depth network whose output is already metric. NeuroLocMap, by contrast, is monocular plus IMU with no external scale reference, and Supplementary Section S.1 shows that on

real KITTI imagery the global scale is not recoverable from motion; the raw RTE (94.7/101.1%) is dominated by this single global offset (the trajectory spans 1.9× the GT length) and is therefore not comparable to scale-consistent systems. Second, even under the most favorable monocular protocol — a 7-DoF Sim(3) alignment that removes the global scale — the per-frame RTE (72.1/75.2%) remains far above the published figures, and excluding the 50 largest per-frame errors still leaves 38.2/46.9%. The residual is a systematic per-frame scale drift, the per-frame projection of the same unrecoverable global scale (Supplementary Section S.1): a property of the up-to-scale VIO core, not of the loop-closure / scale-recovery layer this paper contributes. Where that contribution is visible is global consistency: on KITTI 07 the loop-closure line reduces the EKF core's aligned ATE from 22.09 m to 18.45 m (−16.5%), and across the four CARLA towns the full system reduces the EKF's ATE by up to 59.3% (Table 3). NeuroLocMap is best read as a loop-closure and global-consistency layer on top of an up-to-scale VIO core, rather than a per-frame odometry competitor to scale-consistent stereo systems.

Comparison with Learning-Based Methods. Direct quantitative comparison with learning-based SLAM methods (DROID-SLAM [43], TartanVO [44], DeepVO [45]) is challenging due to fundamentally different evaluation protocols and training requirements. These methods require large-scale training datasets (e.g., TartanVO trained on the large-scale synthetic TartanAir benchmark), GPU-intensive inference, and domain-specific fine-tuning for deployment. Nevertheless, we provide contextual comparison where data is available:

- **Accuracy**: published values use the official KITTI subsegment drift

rate (trel), a different protocol from our terminal/length drift, so each is reported under its own protocol. DeepVO, trained on real KITTI sequences, reports 3.91% trel on sequence 07; TartanVO, trained purely on synthetic data and transferred to KITTI without fine-tuning, reports 4.32% trel on the same sequence. NeuroLocMap, with its learned components trained once on simulation and deployed zero-shot, attains 9.71% terminal/length drift on KITTI 07 (2.9–11.7% across the five test scenarios, Table 3).

- **Computational Efficiency**: DROID-SLAM achieves state-of-the-art accuracy but runs at 4–5 FPS (slower than NeuroLocMap's ∼31 FPS), limiting real-time deployment on embedded platforms.

- **Interpretability**: NeuroLocMap's bio-inspired components (grid cells, HDC, experience map) offer explicit spatial representations that can be visualized and debugged, unlike black-box deep networks.

- **Deployment Protocol**: NeuroLocMap's learned components are trained once on simulation and deployed zero-shot on all test sequences, including the real-world one, while learning methods often require per-domain adaptation.

Our approach targets a different design point: interpretable, train-once real-time localization and mapping for resource-constrained platforms where learning-based methods may not be feasible. For applications requiring maximum accuracy with ample computational resources and training data, learning-based methods remain superior.

Limitations and Future Work. While NeuroLocMap demonstrates strong performance across diverse scenarios, several limitations warrant discussion:

- **Sim-to-Real Scale Gap**: the scale head transfers zero-shot within simulation (including the held-out town) but not yet to real imagery (Supplementary Section S.1); closing this gap — e.g., with scale supervision from sensors available on real vehicles — is the main outstanding work item.

- **4-DoF Constraint**: The current height-discretized orientation representation suits ground vehicles but may limit applicability to fully 6-DoF scenarios with aggressive 3D maneuvers. Future work will extend to spherical orientation encoding for aerial platforms.

- **Template Management**: The densest map (KITTI 07: 964 templates, $\sim$1.4 per meter) shows that on visually non-repeating scenes template creation is not yet sufficiently selective. Incorporating visual saliency or geometric verification could reduce false template creation and bound map size.

- **Loop Verification**: Accepted loops rely on embedding gates only; Town07 shows the residual cost of a marginally wrong re-anchor. Geometric consistency checks on accepted loops could further harden the gate chain.

- **Online Parameter Adaptation**: The framework uses fixed thresholds across scenarios (single global gate set, $\tau = 0.94$). Learning-based

or Bayesian approaches to automatically tune parameters based on environment characteristics could improve robustness.

5. Conclusion

With the increasing demand for robust and interpretable localization systems in resource-constrained environments, achieving biologically-plausible spatial representations has become essential for autonomous navigation and mobile robotics. This study proposes NeuroLocMap, a brain-inspired 4-DoF localization-mapping framework that integrates computational models of 3D grid cells and multilayered head direction cells with dual-stream visual processing and complementary vestibular-visual fusion. We describe the bio-inspired spatial cell architecture, the dual-stream visual processing strategy inspired by ventral and dorsal pathways, and the lightweight complementary fusion mechanism for multi-sensory integration. Experimental results across five diverse scenarios (CARLA Town 05 / 03 / 07 / 10HD, KITTI 07) verify that NeuroLocMap's full system beats the 10-state EKF visual-inertial baseline on *all five scenarios under both evaluation calibers* — by +16.5–59.3% under the shape-only full-Sim(3) refit and by +1.5–33.9% under the rigid SE(3) alignment, in which a wrong metric scale cannot be absorbed by the aligner — and outperforms monocular VO by +33.7–68.8%, while maintaining real-time performance at ∼31 FPS. On a dedicated closed-loop benchmark, its experience-map re-anchoring recovers 9× more of the accumulated global drift than the EKF baseline (44.24 m vs 398.05 m anchored Sim(3)). The framework increases the visual template count from ∼5 templates (typical in RatSLAM) to up to 964 templates (over 192× improvement), enabling finer-

grained place discrimination for robust loop closure detection. Its learned components — the monocular scale-correction head and the loop-closure embeddings — are trained once on simulation and deployed zero-shot on every test sequence, including the real-world one; the loop line transfers directly, while the scale head's current sim-to-real gap is reported honestly. Future work will extend the framework to full 6-DoF pose representation, close the sim-to-real scale gap, and deploy on neuromorphic hardware to progressively realize interpretable localization systems capable of real-time operation on embedded platforms.

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Code availability

The NeuroLocMap code is available at: `https://github.com/OpenHUTB/carla-pedestrians`

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Haidong Wang: Conceptualization, Methodology, Software, Writing – original draft. **Rongbo Yang**: Data curation, Validation, Writing – review & editing.

Data availability

The CARLA datasets were collected with the open-source CARLA simulator (v0.9.16) using the collection code included in the public repository below; the KITTI odometry benchmark is publicly available at `https://www.cvlibs.net/datasets/kitti/`. All processed sequences, trained model weights, and evaluation scripts are available at `https://github.com/OpenHUTB/carla-pedestrians`.

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